DuckRabbit: Typed Multimodal Illusion Generator

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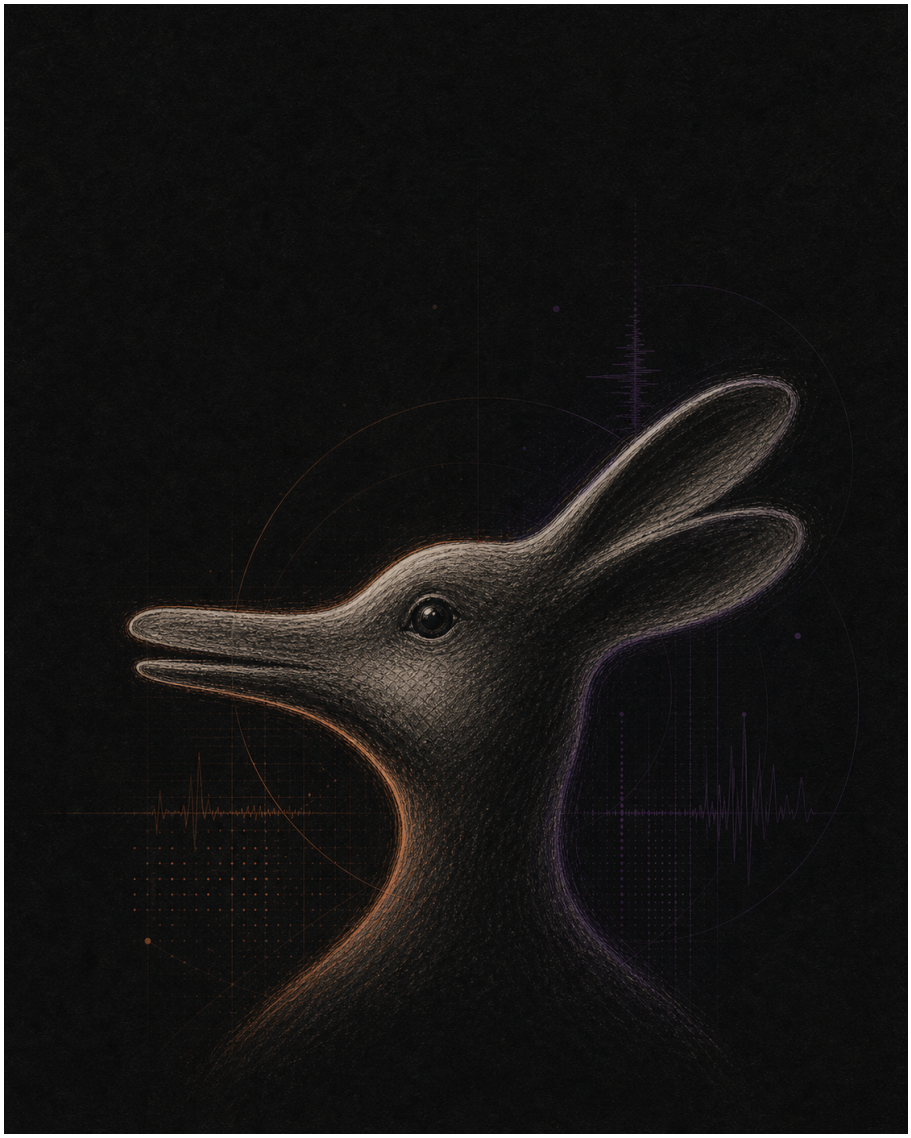
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1 Abstract

DuckRabbit is typed, deterministic research software for constructing reproducible visual, auditory, temporal, and audio-visual stimulus families. It is published and citable at the DuckRabbit public GitHub repository and released under the MIT License.

Its basic unit is an immutable request containing an illusion identifier, validated parameters, a seed, and an encoding specification. The request yields a canonical artifact, objective media measurements, and a versioned provenance manifest before any delivery codec is selected. This separation makes the stimulus a testable computational object while reserving claims about perception for controlled observer protocols.

The release situates its engineering choices within a deliberately broad historical foundation spanning Greek, Arabic/Islamicate, Chinese, and early-modern work on optics, visual inference, and cross-sensory knowledge, alongside later psychophysics and illusion research. These traditions motivate questions about construction, observation, and evidence; they are contextual precedents, not evidence that a generated file reproduces an ancient, early-modern, or clinical observation.

Version 0.5.0 contains 17 implemented generators and 18 catalog entries, supported by 36 source records and 18 evidence records in a checked-in audit snapshot. The package generates 15 publication figures and 10 machine-derived tables from the live registry. It also provides a typed observer-study harness and a transparent synthetic diagnostic: a hand-specified feature observer with serialized weights, temperature, analytic calibration, and `human_data=false`. No participant data are bundled; synthetic model output is explicitly nonhuman.

DuckRabbit verifies physical stimulus properties, media round trips, hashes, timing, and declared provenance. It does not infer a universal percept, effect size, or cross-device perceptual equivalence from a generated artifact. Its contribution is a reproducibility and epistemic boundary: source-backed family descriptions, deterministic media facts, and future observer hypotheses remain different typed records rather than being collapsed into one claim. DuckRabbit is software for reproducible stimulus construction and audit, not a claim that a generated file alone produces a universal perceptual effect.

2 Introduction

An illusion stimulus is not the same thing as an illusion report. A stimulus can be physically specified, generated, encoded, and inspected without establishing what every observer will see or hear. This distinction is central to reproducible cognitive science: the physical manipulation must be auditable, while observer-level effects require a task, calibrated presentation conditions, randomization, and data.

Visual illusion classifications are useful maps rather than exhaustive or universally agreed ontologies [Gregory, 1997]. Auditory and audiovisual families add spectral structure, channel separation, temporal coincidence, and spatial alignment. The Sound-Induced Flash literature illustrates the boundary particularly clearly: the physical contrast is a flash paired with different beep counts, while the perceptual claim requires a listener, timing, and a response task [Hirst et al., 2020, Shams et al., 2000].

DuckRabbit therefore treats an illusion as a typed stimulus-construction problem. Each catalog entry declares modality, mechanism, perceptual signature, cognitive process, requirements, evidence status, output kind, and implementation status. The taxonomy is orthogonal and explicitly provisional; it is an engineering index linked to sources, not a replacement for domain-specific theory.

The release also treats the package as research software rather than as an unannotated collection of media files. Reproducible computational research depends on preserving the code, inputs, environment assumptions, and execution path needed to recreate a result [Sandve et al., 2013, Wilson et al., 2014]. The FAIR literature extends that responsibility to algorithms, tools, and workflows, while software-citation guidance emphasizes identifying the exact software object and version used [Wilkinson et al., 2016, Lamprecht et al., 2020, Smith et al., 2016]. DuckRabbit applies these principles narrowly: it makes the construction pipeline and its generated artifacts citable and inspectable, but it does not treat metadata quality as evidence of a perceptual effect.

The package is organized around three falsifiable engineering hypotheses:

1. Identical typed requests produce identical canonical arrays and canonical digests.
2. Encoders and decoders preserve declared media facts within an explicit format-specific tolerance, or verification fails.
3. Parameter manipulations change measurable physical stimulus properties; any observer-level interpretation remains a preregistered hypothesis until data exist.

This boundary has a nineteenth-century scientific lineage. Fechner's *Elemente der Psychophysik* formalized the problem of relating controlled stimulus differences to measured sensation, while Wheatstone's binocular-vision experiments made the distinction between the physical images delivered to two eyes and the resulting depth interpretation experimentally explicit [Fechner, 1860, Wheatstone, 1838]. Helmholtz's physiological optics then integrated measurement of the eye, visual geometry, and theories of perceptual inference [Helmholtz, 1867]. DuckRabbit does not reproduce those historical experiments; it inherits their methodological lesson that stimulus conditions and observer inferences must be represented as distinct objects.

That lineage has deeper and less geographically narrow roots. Ptolemy's ancient *Optics*, Ibn al-Haytham's medieval Arabic optics, and the Mohist Canon's early Chinese discussions of optics and mechanics show that questions about image formation, geometry, knowledge, and evidence were developed across multiple intellectual traditions [Ptolemy, 1996, al Haytham, 1989, Schemmel and Boltz, 2022, Dai, 2015]. Early-modern discussions sharpened the distinction between a physical presentation and an inferred percept: Kircher documented optical display, Berkeley analyzed learned spatial inference, and the Molyneux–Cheselden record made cross-sensory transfer and observer testimony explicit problems [Kircher, 1646, Berkeley, 1709, Degenaar and Lokhorst, 2020, Cheselden, 1728]. DuckRabbit cites these sources as historical and methodological precedents, not as evidence that its generated files reproduce ancient, early-modern, or clinical observations.

The present catalog contains 18 entries, of which 17 are implemented, 0 are planned, and 1 require external fixtures. This status vocabulary makes breadth visible without claiming that a finite package covers all known phenomena.

3 Methodology

3.1 Request and canonical artifact

DuckRabbit starts from a typed request

$$q = (i, \theta, s, e) \tag{1}$$

where i is an illusion identifier, θ is an immutable parameter object, s is a deterministic seed, and e is an optional encoding request. A registered generator maps the request to a canonical artifact:

$$A = G_i(\theta; s), \qquad A \in \{I, X, V, AV\} \tag{2}$$

The four artifact types are an image frame I, audio buffer X, video sequence V, and audiovisual timeline AV. The generator core is independent of Pillow, WAV containers, GIF, MP4, and ffmpeg. Each artifact retains its shape, dtype, units, clock, channels, and timing offsets.

3.2 Typed parameter domains

Value objects validate finite ranges at construction time. Image parameters include dimensions, color mode, luminance bounds, grayscale levels, and quantization levels. Audio parameters include frequency, phase, amplitude, duration, envelope, sample rate, channel count, and PCM depth. Video parameters include dimensions, rational frame rate, frame count, and temporal offsets. Audiovisual parameters compose audio and video clocks with declared synchronization and spatial discrepancies.

The canonical sample count and frame timestamps are explicit. For audio duration T and sample rate f_s, N is rounded once at construction. For a video with frame rate f_r, frame index k is timestamped on the presentation clock. The audio-minus-video sign convention is explicit:

$$N = \mathrm{round}(f_s T), \qquad t_k = k/f_r, \qquad \Delta_{AV} = t_{\mathrm{audio}} - t_{\mathrm{video}} \tag{3}$$

The package rejects non-finite values, invalid shapes, out-of-range samples, aliasing frequencies, incompatible channel layouts, impossible frame rates, and contradictory encoding requests.

3.3 Canonical serialization and provenance

Canonical arrays are contiguous little-endian float32 buffers with explicit metadata. Serialization includes the array values, shape, clock, and units; metadata are part of the identity rather than an informal annotation:

$$c(A) = \mathrm{Serialize}_{\mathrm{LE,float32}}(A, \mathrm{shape}, \mathrm{clock}, \mathrm{units}), \qquad h_c(A) = H(c(A)) \tag{4}$$

The canonical digest is independent of PNG, WAV, GIF, MP4, and NPZ delivery choices. The v2 manifest records the parameter schema, serialized parameters, taxonomy, evidence boundary, canonical facts, objective metrics, encoding profile, backend identity, decoded inspection, and verification status. A v1 reader remains available and marks upconverted records unverified until the encoded file is re-inspected.

This separation follows a reproducibility principle from computational research: a result is not made reproducible merely by publishing a final file; the executable inputs, transformation steps, versioned software, and provenance must remain identifiable [Sandve et al., 2013, Wilson et al., 2014]. The manifest therefore records the construction identity and delivery identity separately. A downstream user can cite the software version and regenerate the canonical object, or cite a particular encoded artifact when the delivery file itself is the relevant research object [Smith et al., 2016].

The distinction also follows the older psychophysical problem of relating a controlled physical increment to a measured sensation. Fechner's 1860 treatise made measurement, stimulus control, and response comparison explicit parts of the scientific object [Fechner, 1860]. DuckRabbit adopts the stimulus-control and provenance aspects of that tradition, but it does not pretend that a canonical digest or media metric is a sensation measurement: observer data still require a task, presentation conditions, and a registered analysis.

3.4 Encoding and verification

For an encoding request e, the delivered file F and decoded inspection I are:

$$F = E_e(A), \qquad I = D(F), \qquad V(F, M) \in \{\text{pass}, \text{fail}\} \tag{5}$$

The manifest M binds A, F, and I through hashes, typed summaries, and format-specific tolerances. PNG, GIF, WAV, and NPZ use local deterministic adapters. MP4 and muxed audiovisual output use an optional ffmpeg backend. Missing backends are reported as capability errors. Writes are atomic, parent directories are created, overwrite behavior is explicit, and a conflicting format argument is rejected.

3.5 Objective metric definitions

`MetricRecord` and `MetricSuite` carry a metric name, value, unit, computation version, source artifact, tolerance, and claim level. Luminance statistics, quantization levels, RMS and peak amplitude, spectral summaries, frame deltas, stream durations, and synchronization offsets are physical or decoded-media facts. They are not observer responses and cannot by themselves establish an illusion effect. For canonical image samples $y_1, \ldots, y_n$, audio samples $x_1, \ldots, x_n$, and adjacent video frames Y_k, Y_{k+1}, the principal metrics are:

$$\mu_Y = \frac{1}{n}\sum_{j=1}^{n} y_j, \qquad \sigma_Y = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(y_j - \mu_Y)^2}, \qquad RMS_X = \sqrt{\frac{1}{n}\sum_{j=1}^{n} x_j^2} \tag{6}$$

$$D_k = \frac{1}{|Y|}\sum_{p}|Y_{k+1}(p) - Y_k(p)|, \qquad \text{centroid}(X) = \frac{\sum_f f\, M_X(f)}{\max(\sum_f M_X(f), 10^{-12})} \tag{7}$$

Here $M_X(f) = C^{-1}\sum_{c=1}^{C} |\,\text{rFFT}(X_c)(f)|$ is the mean of the channel-wise magnitudes of the unwindowed, one-sided real FFT, including DC and the Nyquist bin when present. It is not a power or energy spectrum, nor the FFT of a channel-averaged waveform. Frequencies are in Hz at the canonical sample rate; the same magnitude weights define the spectral bandwidth. The denominator floor yields a zero centroid for silence. The illustrative audio atlas uses separately disclosed display-band aggregation; it does not replace this full-resolution metric computation.

The level count is the cardinality of the finite-value set after canonical quantization; an empty or non-finite artifact is invalid rather than assigned a default statistic. Every metric record carries units and a deterministic computation version. Observer interpretations are represented separately by the typed observer protocol and claim levels.

For a future observer study, the estimand is defined by the protocol rather than by the generator itself:

$$\psi = \mathbb{E}[Y \mid \text{condition},\ \text{protocol}] \tag{8}$$

This symbol names a planned analysis target only. DuckRabbit ships no human responses, fitted coefficients, or validated observer-level estimate.

3.5.1 Synthetic psychophysics boundary

DuckRabbit v0.5.0 adds a transparent synthetic-psychophysics diagnostic for method orchestration. It is a hand-specified feature observer, not a pretrained vision model and not an empirical psychophysics substitute. Let $\phi(A)$ be the bounded feature vector extracted from a canonical artifact, w the code-owned feature weights, and $\tau > 0$ the declared temperature:

$$p_M(c \mid A_r, A_c) = \sigma\left(\frac{w^\top \phi(A_c) - w^\top \phi(A_r)}{\tau}\right) \tag{9}$$

The model identity, version, preprocessing features, weights, temperature, seed, calibration statement, and the `human_data=false` flag are serialized with each prediction. The diagnostic is useful for checking that typed parameter sweeps produce deterministic, inspectable model outputs and for exercising analysis orchestration end to end. It does not estimate a human psychometric function, establish a sensitivity threshold, or support an observer-level claim. Human psychophysics remains future work requiring a preregistered task, calibrated display/playback, participant data, and an analysis contract.

4 Results

4.1 Generated catalog, metrics, and outputs

The live registry reports 17 implemented generators, 18 catalog entries, 18 evidence records, and 36 source records in the checked-in scholarship audit snapshot. The complete catalog is generated in Appendix A rather than copied into prose. The companion figure keeps the categorical registry, source namespace, and implementation statuses visible at a glance.

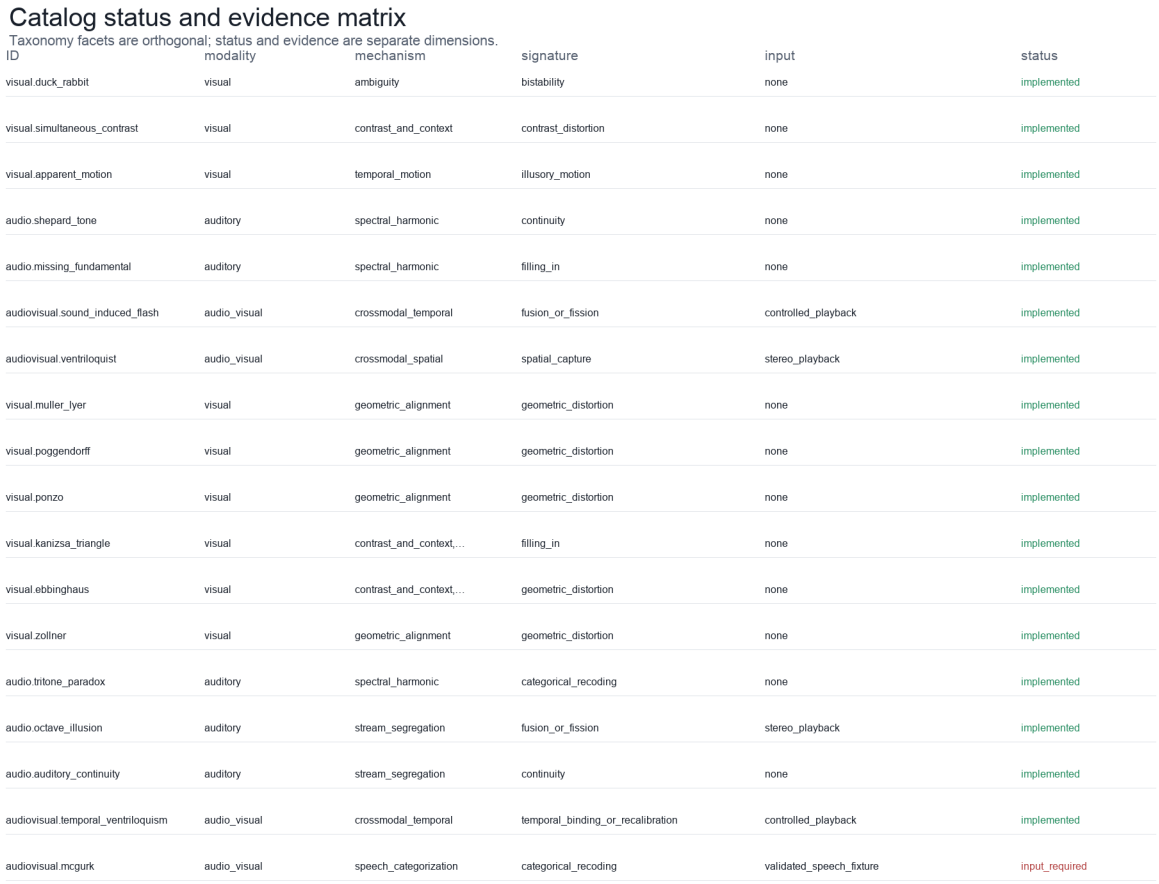


Figure 1: A matrix lists all catalog entries with modality, mechanism, signature, input requirement, and implementation status.

What this figure shows: A matrix lists all catalog entries with modality, mechanism, signature, input requirement, and implementation status. The live catalog matrix contains 18 entries and keeps modality, mechanism, perceptual signature, input requirement, evidence status, and implementation status as separate facets. Source data output/data/catalog_matrix.json preserve the exact taxonomy snapshot and its source-reference namespace; status is not a proxy for perceptual validation. Controls: live taxonomy registry; evidence matrix snapshot; seed not applicable to the catalog diagram. Objective facts: 18 catalog rows; categorical facets and statuses; no observer-level quantity is applicable. Claim level: source_supported. Source data: output/data/catalog_matrix.json (SHA-256 digest recorded in the figure registry). Evidence lineage: gregory1997visual, hirst2020sound, bruns2019ventriloquist. Limitations: Taxonomy facets are engineering classifications and remain provisional where the literature supports competing accounts. Boundary: Catalog coverage is bounded to this registered package and does not enumerate all known illusions.

The complete catalog matrix fig. 1 is tabulated in the standalone Appendix A sec. 10 and in tbl. 10. The corresponding machine-readable evidence matrix records the source role, exact supported claim, engineering departure, and limitation for every entry. Keeping the table in an appendix gives the main results narrative room to explain the contract without

turning a mutable registry snapshot into a prose claim.

4.2 Typed domains and delivery profiles

Table 1: Typed parameter domains and units.

Type	Unit	Validated domain	Role
PixelDimension	pixels	8–4096	image/video width and height
GrayscaleLevels	levels	2–256	representable grayscale levels
QuantizationLevels	levels	2–256	output quantization buckets
FrequencyHz	Hz	> 0 and finite	oscillator and harmonic frequencies
SampleRate	samples/s	1,000–384,000	audio sampling clock
FrameRate	frames/s	1–240	video presentation clock
SyncOffsetMs	ms	−10,000–10,000	audio relative to video
SpatialOffset	normalized	−1–1	declared crossmodal spatial discrepancy

The parameter contracts are summarized in tbl. 1. They constrain physical construction and serialization; they are not psychophysical scales.

Table 2: Encoding profiles and backend capabilities.

Format	Backend	Artifact	Profile	Status
PNG	Pillow	image	lossless uint8 raster	available
WAV	python wave	audio	PCM 8/16/24/32-bit	available
GIF	Pillow	video	palette animation with declared duration	available
NPZ	NumPy	canonical artifact	exact little-endian float32 archive	available
MP4	ffmpeg	video/audiovisual	H.264 or muxed MP4; decode inspected	available

The delivery profiles are summarized in tbl. 2. Backend availability is environment-dependent and is recorded at generation time.

The catalog distinguishes a physical construction from an observer claim. For example, the sound-induced-flash generator establishes one flash and one or two beep events on a shared clock; it does not establish a reported flash count. Similarly, the temporal-ventriloquism generator exposes a declared timing offset, while the literature reports observer-dependent temporal judgments under specific tasks and timing conditions [Vroomen and de Gelder, 2004].

4.3 Pipeline and provenance

What this figure shows: Five labeled stages connect a typed request to a deterministic canonical artifact, encoded media, decoded inspection, and a verification manifest. This schematic separates a typed request from the canonical artifact it generates, the optional delivery encoder, decoded inspection, and the manifest-level verification decision. Arrows indicate data and provenance dependencies rather than a causal model of perception; the seed is 0 and the complete stage record is in source data output/data/architecture.json. Controls: typed request q=(i,θ,s,e) with seed s=0; default generator and delivery-independent canonicalization. Objective facts: no unit-bearing objective quantity is applicable to this architecture diagram; stage identities and provenance fields are recorded in the sidecar. Claim level: canonical_stimulus. Source data: output/data/architecture.json (SHA-256 digest recorded in the figure registry). Evidence lineage: formalism:eq:typed_request, formalism:eq:canonical_generation, formalism:eq:encoding_verification. Limitations: The schematic abstracts implementation boundaries and does not represent an observer study or a causal theory of perception. Boundary: The pipeline verifies reproducible media facts, not what a person experiences.

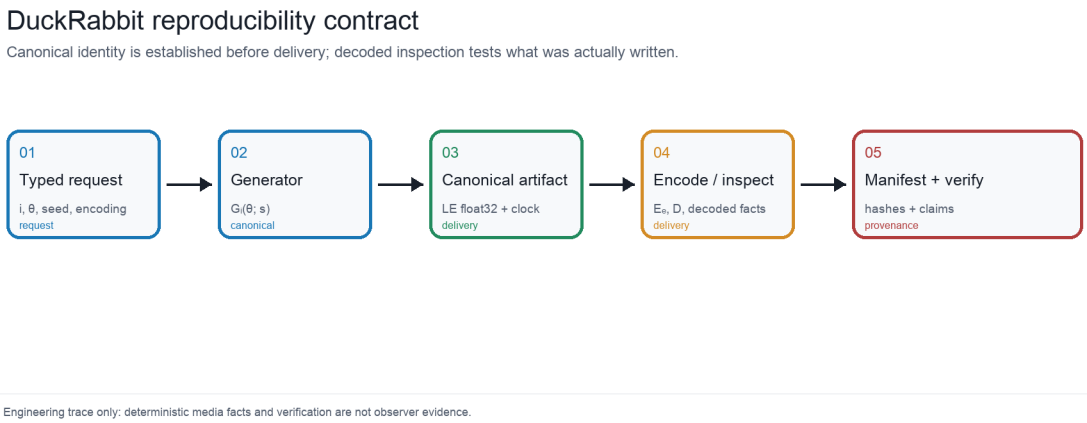


Figure 2: Five labeled stages connect a typed request to a deterministic canonical artifact, encoded media, decoded inspection, and a verification manifest.

The generated architecture figure fig. 2 shows the separation between request, canonical artifact, delivery adapter, decoded inspection, and manifest. The canonical digest is the identity of the in-memory stimulus; an encoded hash is the identity of the delivered file. The difference is intentional.

4.4 Visual constructions and parameter sweeps

What this figure shows: A 9-panel gallery shows every currently implemented visual catalog entry, with stable IDs and physical raster summaries: Duck-rabbit ambiguous figure, Simultaneous contrast stimulus, Apparent-motion frame sequence, Müller-Lyer geometric illusion, Poggendorff geometric illusion, Ponzo perspective illusion, Kanizsa illusory-contour triangle, Ebbinghaus context-size illusion, Zöllner orientation illusion. This complete gallery presents one deterministic representative from each of the 9 currently implemented visual catalog entries: Duck-rabbit ambiguous figure, Simultaneous contrast stimulus, Apparent-motion frame sequence, Müller-Lyer geometric illusion, Poggendorff geometric illusion, Ponzo perspective illusion, Kanizsa illusory-contour triangle, Ebbinghaus context-size illusion, Zöllner orientation illusion. The panel therefore covers the package's available visual families—ambiguous figure, contrast, apparent motion, geometric context, illusory contour, contextual size, and orientation-related constructions—using typed defaults at seed 0. For the temporally expressed apparent-motion entry, the displayed tile is its first frame and the source data preserve the full sequence. The source data sidecar output/data/visual_panel.json records the stable ID, generator parameters, canonical SHA-256 digest, Rec. 709 or grayscale luminance statistics, raster-level counts, and parameter boundary for every tile. Controls: one default typed parameter object per implemented visual entry; seed s=0; dimensions, luminance bounds, grayscale, and quantization controls remain in each generator schema; temporal entries display their first frame but retain sequence metrics. Objective facts: per tile: mean and standard-deviation relative luminance, unique raster levels, and canonical SHA-256 digest; temporal entries additionally retain frame count, frame rate, and frame deltas; definitions and units are in the source data JSON. Claim level: canonical_stimulus. Source data: output/data/visual_panel.json (SHA-256 digest recorded in the figure registry). Evidence lineage: gregory1997visual, brugger1999duckrabbit, howe2005muller, morgan1999poggendorff, fisher1967ponzo, yildiz2022ponzo, kanizsa1976contours, mruczek2015ebbinghaus, earle1995zollner, wertheimer1912motion, sekuler1996wertheimer. Limitations: The gallery is complete for implemented visual entries in this package, not a complete survey of visual illusions; literature families, historical displays, and DuckRabbit rasters are not pixel-identical by default. Viewing scale, display calibration, viewing distance, and observer conditions can change the relevance of a construction; the gallery does not measure an observer's perceptual report. Boundary: The figure establishes coverage of the package's implemented visual constructions and their deterministic raster facts. It does not establish that any viewer will perceive the named signature, nor that the set is exhaustive of the visual-illusion literature.

The visual panel fig. 3 is deliberately a coverage figure: it contains one deterministic representative for every currently implemented visual entry, including apparent motion and Zöllner in addition to the static ambiguous-figure, contrast, geometric-context, illusory-contour, and contextual- size families. Equalities, masks, line geometry, luminance bounds, and temporal positions are properties of the typed rasters or sequences. They are not observer scores, and the gallery is not an exhaustive ontology of visual illusions.

What this figure shows: A labeled grid crosses five duck–rabbit blend weights with five grayscale and quantization settings, with physical metrics recorded for each cell. This 5×5 sweep varies duck–rabbit blend weight across rows and jointly varies

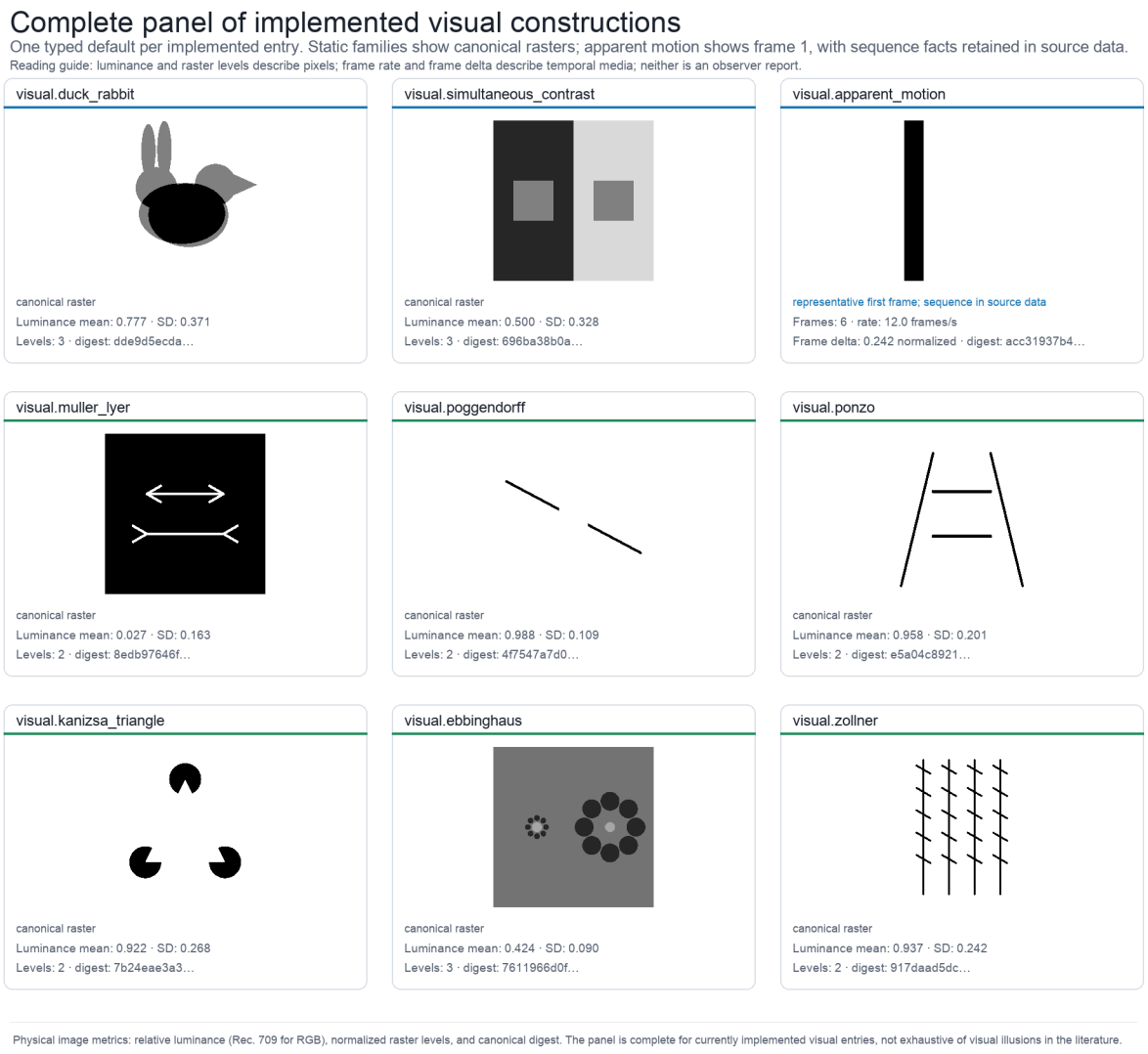


Figure 3: A 9-panel gallery shows every currently implemented visual catalog entry, with stable IDs and physical raster summaries: Duck-rabbit ambiguous figure, Simultaneous contrast stimulus, Apparent-motion frame sequence, Müller-Lyer geometric illusion, Poggendorff geometric illusion, Ponzo perspective illusion, Kanizsa illusory-contour triangle, Ebbinghaus context-size illusion, Zöllner orientation illusion.

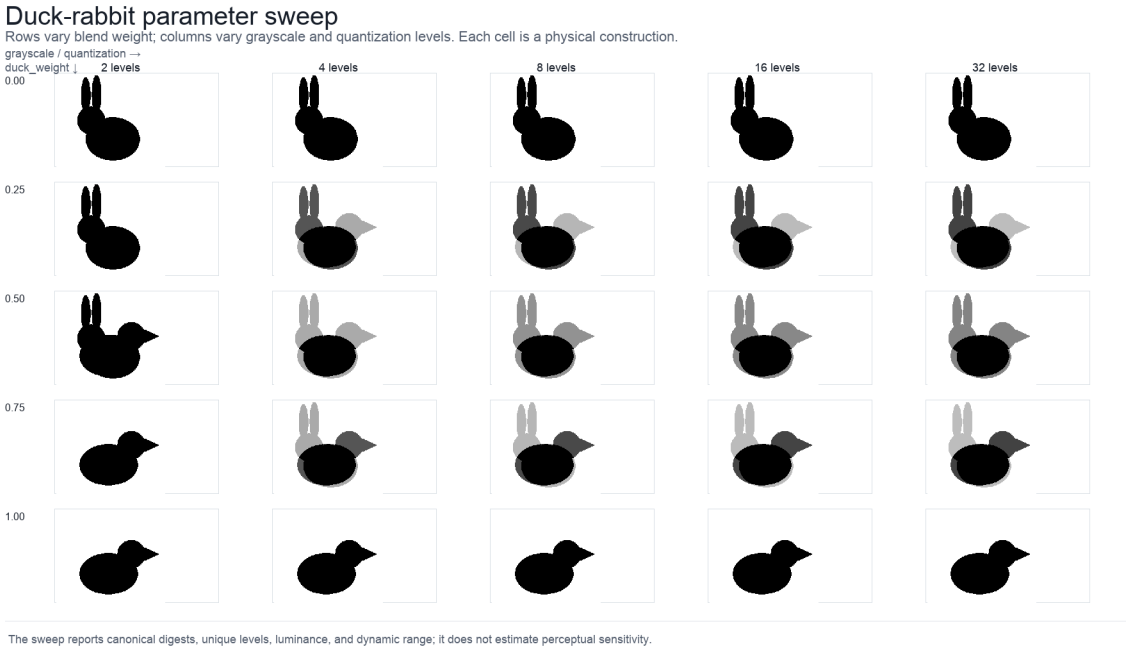


Figure 4: A labeled grid crosses five duck–rabbit blend weights with five grayscale and quantization settings, with physical metrics recorded for each cell.

grayscale and quantization levels across columns. Each cell is regenerated from an immutable typed parameter object at seed 0; source data output/data/visual_sweep.json records canonical digests, unique-level counts, normalized luminance, and effective dynamic range for every condition. Controls: duck_weight $\in \{0,.25,.5,.75,1\}$; grayscale=quantization $\in \{2,4,8,16,32\}$; seed s=0. Objective facts: 25 canonical rasters; unique levels, normalized mean luminance, and effective dynamic range per cell. Claim level: physical_metric. Source data: output/data/visual_sweep.json (SHA-256 digest recorded in the figure registry). Evidence lineage: brugger1999duckrabbit, gregory1997visual. Limitations: The grid is a sensitivity of the construction parameters, not a psychophysical sensitivity curve or a validated observer effect. Boundary: The chosen grid is an engineering sampling of the parameter domain and does not imply an optimal or perceptually uniform spacing.

The sweep fig. 4 varies duck-weight from 0 to 1 and jointly varies representable grayscale and quantization levels. The source data preserve canonical digests and unique-value counts; no perceptual score is assigned to a sweep cell.

4.5 Temporal and auditory constructions

What this figure shows: Six ordered frames are labeled with timestamps and paired with a line plot of adjacent-frame normalized pixel differences. The frame strip exposes the order and timestamps of the apparent-motion construction, while the lower trace reports adjacent-frame mean absolute pixel differences. Source data output/data/temporal_sequence.json preserve frame count, frame rate, reduced rational timebase, timestamps in seconds, and the nonzero transition series; seed 0 identifies the generated sequence. Controls: visual.apparent_motion default typed parameters; frame rate and frame count from the canonical sequence; seed s=0. Objective facts: timestamps in seconds, frame rate in frames/s, frame count, rational timebase, and adjacent-frame normalized pixel difference. Claim level: physical_metric. Source data: output/data/temporal_sequence.json (SHA-256 digest recorded in the figure registry). Evidence lineage: wertheimer1912motion, sekuler1996wertheimer. Limitations: Frame succession and frame deltas are physical file properties; playback timing, display persistence, and motion reports require a controlled observer task. Boundary: The figure documents temporal structure rather than the presence, direction, or strength of perceived motion.

The temporal figure fig. 5 reports frame succession, rational frame rate, and frame-to-frame mean absolute differences. These facts establish temporal structure in the file, not the presence or strength of perceived motion.

What this figure shows: Five labeled rows show canonical audio waveforms, relative one-sided spectral bars, RMS and peak in normalized amplitude, and spectral summaries in hertz. Five implemented auditory constructions—Shepard tone, missing fundamental, tritone paradox, octave illusion, and auditory continuity—are shown as canonical waveforms with compact one-sided FFT summaries. Source data output/data/audio_signals.json records sample rate, channel layout, RMS, peak, spectral centroid, bandwidth, crest factor, canonical digest, and the declared display-bin convention; no

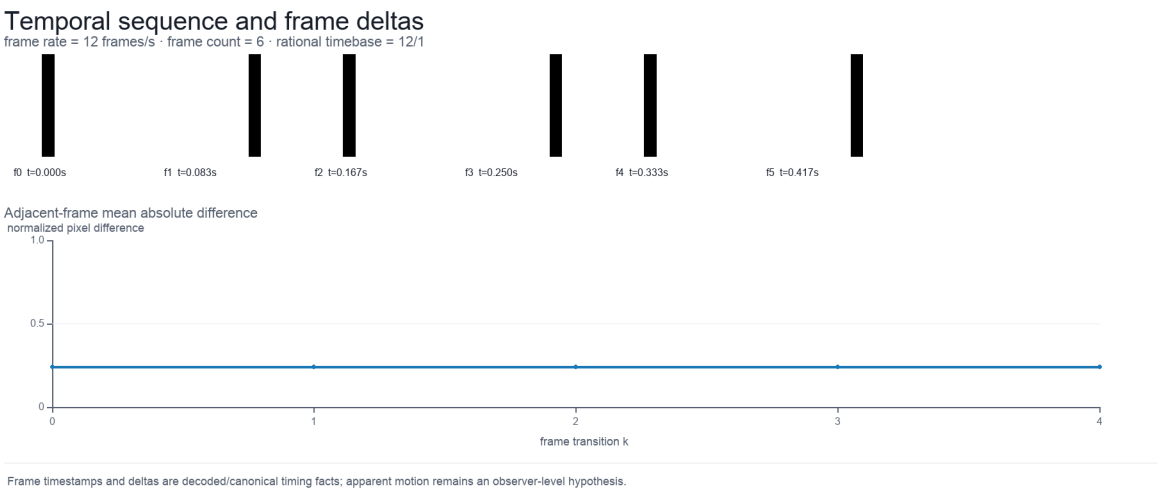


Figure 5: Six ordered frames are labeled with timestamps and paired with a line plot of adjacent-frame normalized pixel differences.

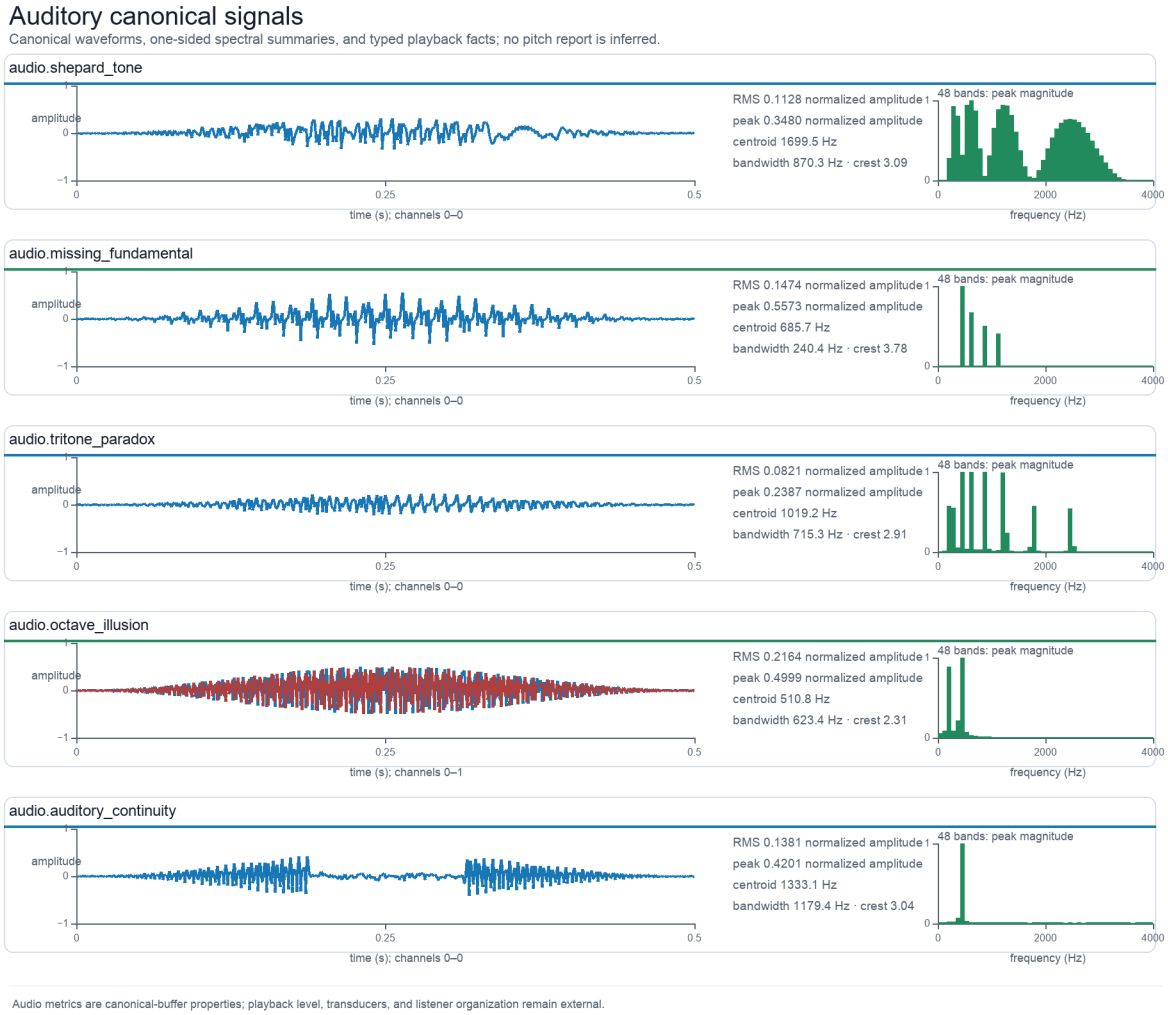


Figure 6: Five labeled rows show canonical audio waveforms, relative one-sided spectral bars, RMS and peak in normalized amplitude, and spectral summaries in hertz.

pitch or stream report is inferred. Controls: default typed audio parameters; per-generator channel layout and sample rate; seed s=0; one-sided FFT display with 48 sampled bins. Objective facts: RMS and peak in normalized amplitude; spectral centroid and bandwidth in Hz; sample rate in samples/s; channel layout and canonical SHA-256 digest. Claim level: physical_metric. Source data: output/data/audio_signals.json (SHA-256 digest recorded in the figure registry). Evidence lineage: shepard1984scale, zatorre2005missing, deutsch1986tritone, repp1997tritone, deutsch1974octave, warren1970continuity, riecke2011continuity. Limitations: Playback level, transducer response, dichotic separation, masker design, and listener judgments are external to the canonical buffer. Boundary: The figure compares generated signal properties and literature-linked families; it does not establish pitch, continuity, or stream organization for a listener.

The auditory figure fig. 6 shows canonical waveforms and spectral summaries for Shepard, missing-fundamental, tritone, octave, and continuity constructions. RMS, peak, spectral centroid, bandwidth, and channel layout are calculated from the canonical buffer with explicit units and windowing rules. Pitch direction, pitch height, continuity, and stream organization remain observer-level outcomes that depend on playback and task conditions.

4.6 Audiovisual timing and encoding

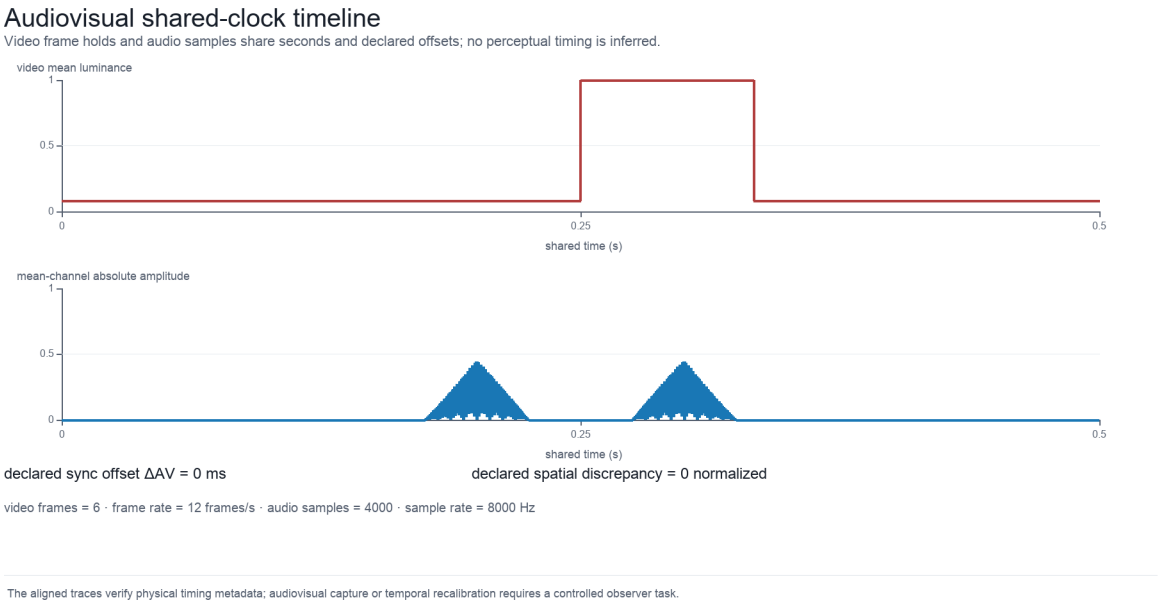


Figure 7: Two labeled traces share a time axis in seconds: video mean luminance and audio absolute amplitude, with signed synchronization and spatial offsets printed below.

What this figure shows: Two labeled traces share a time axis in seconds: video mean luminance and audio absolute amplitude, with signed synchronization and spatial offsets printed below. Video mean luminance and rectified audio amplitude are placed on a common time axis in seconds with video frame holds and actual audio sample timestamps for the sound-induced-flash construction. Source data output/data/audiovisual_timeline.json records the video frame clock, audio sampling clock, event traces, signed audio-minus-video synchronization offset in milliseconds, and normalized spatial discrepancy; alignment is declared by the generator, not estimated from perception. Controls: audiovisual.sound_induced_flash default shared timeline; declared sync offset ΔAV and spatial offset; seed s=0. Objective facts: video frame-hold luminance and mean-channel absolute audio amplitude on shared time in seconds; ΔAV in ms; spatial discrepancy in normalized units; frame and sample clocks in the sidecar. Claim level: physical_metric. Source data: output/data/audiovisual_timeline.json (SHA-256 digest recorded in the figure registry). Evidence lineage: shams2000sifi, hirst2020sound, vroomen2004temporal, hartcherobrien2011temporal. Limitations: Display refresh, audio hardware, event latency, and observer temporal binding are not measured by this figure. Boundary: A shared-clock construction is not evidence that observers bind the events at the declared or any other perceptual time.

The audiovisual timeline fig. 7 displays video luminance and audio amplitude against the shared clock, alongside declared synchronization and spatial offsets. Ventriloquist stimuli require stereo playback and spatially resolved display; temporal binding requires controlled timing and an observer task [Bruns, 2019, Vroomen and de Gelder, 2004].

What this figure shows: A five-row comparison lists format, backend, preserved canonical facts, and the corresponding decoded verification checks. The comparison separates the canonical artifact from delivery containers: PNG, WAV, GIF,

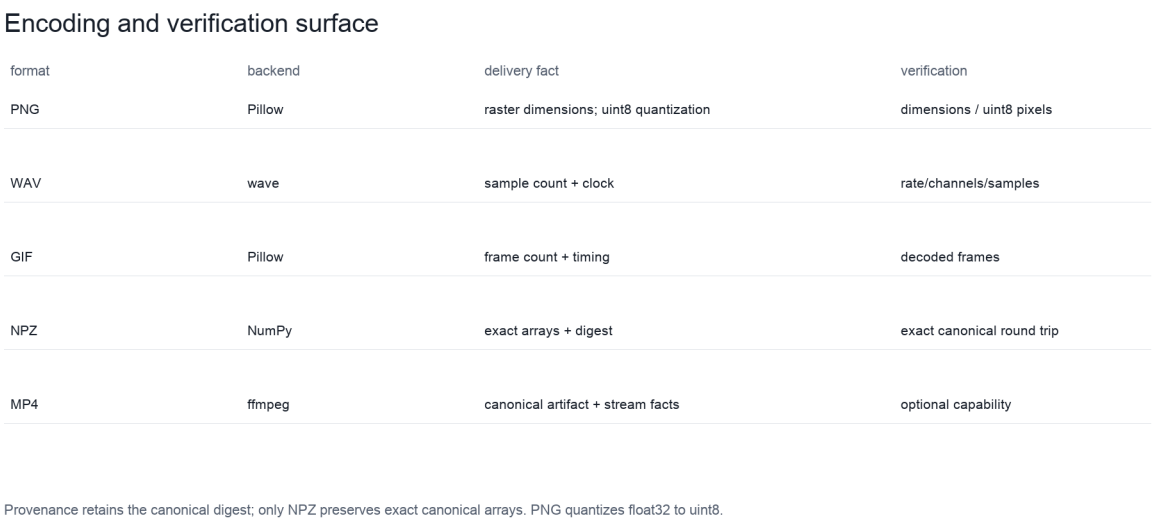


Figure 8: A five-row comparison lists format, backend, preserved canonical facts, and the corresponding decoded verification checks.

NPZ, and optional MP4/muxed MP4. Source data output/data/encoding_verification.json records backend identity, preserved canonical facts, decoded inspection targets, and capability status as observed in the generation environment; the table therefore describes an adapter contract rather than a claim about codec quality. Controls: typed encoding profile selected per artifact; overwrite and backend capability checks; seed inherited from the stimulus request. Objective facts: format availability and verification scope are environment observations; NPZ is exact while other formats are decoded and tolerance-checked. Claim level: encoded_media. Source data: output/data/encoding_verification.json (SHA-256 digest recorded in the figure registry). Evidence lineage: engineering:typed_media_adapters. Limitations: Backend availability, codec versions, container metadata, and lossy tolerances are environment-dependent; playback software can expose additional behavior. Boundary: A successful encode or decode does not validate an observer effect or guarantee cross-device perceptual equivalence.

The encoding matrix fig. 8 makes backend capability and verification scope visible. NPZ is the exact canonical archive; PNG, WAV, GIF, and MP4 are delivery containers whose decoded facts are checked against the manifest.

4.7 Objective metric results

Table 3: Objective metric definitions and epistemic boundary.

Metric	Unit	Definition	Tolerance	Claim level
mean_luminance	normalized_luminance	mean Rec. 709 relative luminance (grayscale is identity)	exact canonical	physical_metric
unique_values	levels	count of distinct canonical image values	exact integer	physical_metric
rms	normalized_amplitude	root-mean-square audio amplitude	finite scalar	physical_metric
peak	normalized_amplitude	maximum absolute audio amplitude	finite scalar	physical_metric
spectral_centroid_hz	Hz	one-sided frequency mean weighted by mean-channel abs(unwindowed rFFT) magnitude, not energy	finite scalar	physical_metric

Metric	Unit	Definition	Tolerance	Claim level
spectral_bandwidth_hz	Hz	spectral standard deviation around the one-sided centroid	finite scalar	physical_metric
crest_factor	ratio	audio peak divided by RMS with zero-RMS guard	finite scalar	physical_metric
mean_temporal_delta	normalized_pixel_difference	mean adjacent-frame absolute difference	finite scalar	physical_metric
sync_offset	ms	declared audio-minus-video offset	typed value	physical_metric

The metric definitions in tbl. 3 are computed from canonical or decoded artifacts and do not encode observer interpretations.

The default duck-rabbit artifact contains 3 unique raster levels and has mean normalized luminance 0.7770. These values are live generated facts, not perceptual effect sizes.

4.8 Observer estimands and verification controls

Table 4: Data-free observer outcomes and preregistered estimands.

Estimand	Response and model	Reference and contrast	Uncertainty
sifi flash count contrast	event count; poisson or ordinal	sifi one beep; two-beep minus one-beep event-count response	95% confidence interval
temporal binding offset contrast	continuous magnitude; linear mixed	temporal sync; reported timing shift per declared sync offset	95% confidence interval

The analysis contract in tbl. 4 specifies estimands and uncertainty procedures without asserting any result.

Table 5: Verification failure modes and negative controls.

Failure mode	Control	Expected result
altered encoded file	recompute encoded SHA-256	fail
stale canonical digest	recompute canonical little-endian float32 digest	fail
incorrect decoded dimensions	compare inspection facts with manifest summary	fail
stream timing mismatch	compare duration/timebase and declared sync offset	fail
unsupported backend	capability probe before encoding	explicit capability error
conflicting format requests	reject format_name/output_spec disagreement	parameter error

The negative controls in tbl. 5 test package integrity and media contracts; they are not null findings about perception.

4.9 Synthetic psychophysics model diagnostic

What this figure shows: A narrow-range line plot shows deterministic model probability across duck_weight values beside the model identity, temperature, no-training-data statement, and human-data boundary. The hand-specified feature observer compares duck_weight variants with a fixed canonical reference. Its probability trace is computed analytically from serialized features, weights, and temperature; source data output/data/synthetic_psychophysics.json records model ID/version, reference and comparison digests, seed, calibration, `training_data=none`, and `human_data=false`. The y-axis is

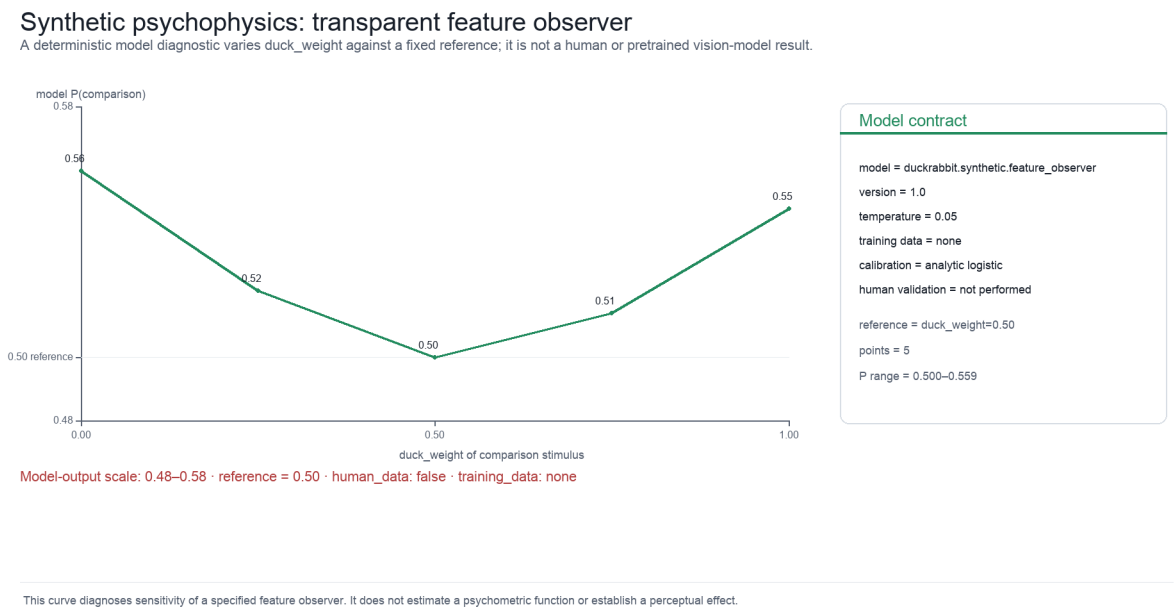


Figure 9: A narrow-range line plot shows deterministic model probability across duck_weight values beside the model identity, temperature, no-training-data statement, and human-data boundary.

deliberately a narrow model-output range rather than a human psychometric scale. Controls: duck_weight comparison against fixed reference; serialized feature weights and temperature; seed s=0. Objective facts: model probability and score delta on the displayed model-output scale; canonical stimulus digests; human_data=false; training_data=none. Claim level: synthetic_model_output. Source data: output/data/synthetic_psychophysics.json (SHA-256 digest recorded in the figure registry). Evidence lineage: observer:preregistered_estimands. Limitations: The model is hand-specified, has no training data, has not been calibrated against observers, and does not estimate a human threshold or effect size. Its probabilities are model outputs, not participant responses or a psychometric function. Boundary: This diagnostic tests deterministic orchestration and metamorphic input-output behavior only; empirical psychophysics remains future work.

The diagnostic fig. 9 evaluates a fully serialized, hand-specified feature observer against a fixed canonical reference while varying `duck_weight`. Its logistic probabilities are model outputs generated from explicit features, weights, temperature, and seed; they are not a pretrained vision-model score, a human psychometric function, an assumed human effect size, or participant data. A future human study would require a preregistered task, calibrated display and playback, consented participants, exclusion and missingness rules, and an analysis contract before any observer claim could be made.

5 Experimental and Computational Setup

Core generation uses Python, NumPy, Pillow, and the standard library. The default seed is 0 and generation is offline. The generators are analytic in their typed parameters: the seed is recorded in manifests and study plans but does not yet alter artifact bytes; seed-driven stimulus randomization awaits a stochastic generator. PNG, GIF, WAV, and NPZ are available without ffmpeg; MP4 and muxed audiovisual output require both ffmpeg and ffprobe. Optional capabilities fail explicitly rather than silently changing the requested artifact.

The deterministic validation suite executes real NumPy synthesis and real temporary-file media round trips. It checks finite normalized arrays, typed parameter boundaries, canonical hashes, dimensions, channel layouts, sample rates, frame counts, timebases, duration agreement, synchronization offsets, decoded stream facts, and registry/evidence consistency.

This is a software-validation protocol, not a psychophysical experiment. The tests establish repeatability under the declared software and dependency conditions; they do not establish replicability of an observer effect under a new display, playback chain, population, or task. That distinction is why the release reports both the exact environment contract and the missing observer evidence.

The publication workflow writes 15 scientific PNG figures plus a separately provenance-recorded editorial cover, machine-readable source-data sidecars, a typed figure registry, 10 markdown tables, and a publication report. Each generated figure records its source function, seed, caption, alt text, claim level, and SHA-256 digest. Generated files are disposable; the tracked source is the generator and its tests.

The registry and cover report are independently revalidated after generation. The registry validator rechecks code-owned captions, evidence namespaces, source-data hashes, and figure hashes. The cover validator rechecks the editorial source digest, portrait dimensions, variant set, and delivery hashes. These checks prevent a successfully rendered image from becoming an unexamined publication claim.

The observer harness is a data-free design layer. It creates reproducibly randomized trial records, exports typed study plans, generates synthetic responses for contract tests, and exposes model templates. It does not collect or ship participant data. The publication atlas additionally runs `duckrabbit.synthetic.feature_observer`, an explicitly hand-specified feature model with no training corpus. Its canonical pairwise probabilities are serialized as model output and are useful for end-to-end orchestration tests; they are not a real vision-model estimate, a psychometric function, or an observer result. Human sensitivity is deferred until a calibrated, consented, preregistered study supplies response data and an analysis contract.

6 Reproducibility and Provenance

The reproducibility contract is:

1. construct the same frozen parameter dataclass;
2. use the same generator ID and the same deterministic seed;
3. compare the canonical artifact digest;
4. record the parameter schema, typed parameters, taxonomy, evidence boundary, canonical byte serialization, objective metrics, and encoding request;
5. if a file is written, inspect the decoded media and compare dimensions, channels, rates, frame counts, durations, timing offsets, and encoded hash;
6. retain the v2 manifest and source-data sidecars with the generated figure or table. Note that in this release the generators are analytic in their typed parameters: the request seed s is recorded in manifests and study plans and excluded from the canonical digest, but it does not yet alter artifact bytes; seed-driven stimulus randomization awaits a stochastic generator.

The contract is deliberately stronger than "the file can be downloaded." Research-software guidance treats versioned code, executable procedures, dependencies, and retained inputs as part of the reproducible object [Sandve et al., 2013, Wilson et al., 2014]. FAIR guidance further asks that digital research objects be findable, accessible, interoperable, and reusable, while noting that software has lifecycle and maintenance constraints that are not identical to those of static data [Wilkinson et al., 2016, Lamprecht et al., 2020]. DuckRabbit operationalizes those principles with typed metadata, deterministic hashes, source-data sidecars, explicit licenses, and release gates rather than by implying that a generated media file is a complete scientific replication.

The package reports implemented, planned, input_required as its catalog status vocabulary and currently covers audiovisual, auditory, visual. The evidence layer contains 18 entry records backed by 36 source records, covering 18/18 catalog entries. The publication workflow produces 15 figures and 10 tables from the live registry.

The public software record is the DuckRabbit public GitHub repository, with Daniel Ari Friedman of the Active Inference Institute as author. The release is archived on Zenodo at 10.5281/zenodo.21419693 (concept DOI, always resolves to the latest version), with 10.5281/zenodo.21419694 identifying this v0.5.0 release specifically.

The canonical buffer is the primary identity of a stimulus. Encoded files are delivery artifacts with format-specific tolerances. The observer protocol is a separate evidence layer: a manifest can prove what was presented without proving what an observer experienced.

Synthetic psychophysics is a third, deliberately bounded object: the model is a hand-specified deterministic feature observer; no training data; human_data=false, registered as `duckrabbit.synthetic.feature_observer`. The model version, feature schema, weights, temperature, seed, calibration statement, canonical reference/comparison digests, and `human_data=false` flag are retained in `output/data/synthetic_psychophysics.json`. Re-running this diagnostic checks deterministic model orchestration and stimulus-feature sensitivity; it does not estimate a human observer or replace empirical psychophysics.

The v2 verifier is a relational check rather than a field-presence check. It rejects summaries whose dimensions, dtype, byte count, rates, duration, signal range, frame deltas, or signed audiovisual offset disagree. Public manifest and inspection mappings are recursively frozen after validation, and rational frame rates are reduced before entering the clock contract. The read-only capability probe records whether Pillow, NumPy, ffmpeg, or ffprobe are available in the current environment; unavailable delivery paths remain explicit rather than being silently downgraded.

The minimal regeneration commands are:

```
uv run pytest -q --cov=src/duckrabbit --cov-fail-under=90
uv run python scripts/generate_publication_outputs.py
uv run python scripts/generate_manuscript_variables.py
```

6.1 Runtime-performance reproducibility

The package also provides `scripts/profile_workloads.py` for fixed real catalog, hashing, metric, observer, media, publication, and audit workloads. Source hashes, runtime-input hashes, environment details, and deterministic result signatures accompany the samples. Warm-up and repeated ordinary timing calls are separate from cProfile and allocation-tracing calls. A comparison is invalid when its environment, scenario, or result differs; profiling timings are not estimates of perceptual latency or participant performance.

Python 3.15 supports the profiling workflow, while the portable package retains its declared Python compatibility range. Optional native sampling remains subject to host process-inspection permissions; an unavailable sampler is reported as unavailable, not inferred to have succeeded. The dated engineering review and raw timing receipts live in `docs/performance-review-2026-10-17.md` and `docs/benchmarks/`, outside the scientific evidence matrix.

6.2 Data availability and software citation

No participant records, fitted observer coefficients, or human psychophysics are bundled. The synthetic diagnostic is explicitly model output with `human_data=false` and `training_data=none`. Reuse should cite Daniel Ari Friedman and the release metadata in `CITATION.cff`, together with the source-specific scholarship in `references.bib`. Software-citation principles recommend citing the software object itself, with enough version and identity information to distinguish one release from another [Smith et al., 2016]. The DOI field carries the real, minted concept DOI (`10.5281/zenodo.21419693`) and `doi_status` is `published`; no resolver URL was manufactured before that identifier existed.

7 Scope, Related Work, and Limitations

7.1 Scope and epistemic boundary

DuckRabbit is a typed, reproducible stimulus-construction platform. Its unit of work is an immutable request that yields a canonical image, audio buffer, video sequence, or audiovisual timeline, together with objective measurements and provenance. It is not an exhaustive ontology of all illusions, a perceptual theory, a participant database, or a substitute for a controlled psychophysics experiment. The live registry is therefore a deliberately bounded catalog: it enumerates the families for which this release has both a generator contract and a stated evidence boundary. Appendix A gives the complete current matrix; the absence of a family from that matrix is not a claim that the family is unimportant or unsupported in the wider literature.

The central distinction is between a construction and an observation. A DuckRabbit generator can establish the dimensions, pixel values, sample values, frequency components, frame clock, declared synchronization offset, encoded hash, and decoded media facts of its output. It cannot, from those facts alone, establish what a person sees, hears, counts, localizes, groups, or reports. Every figure caption, evidence record, and claim-ledger entry preserves this boundary. In particular, a literature citation establishes a source-supported statement about a stimulus family or result under that source's conditions; it does not certify pixel- or task-identical replication by a DuckRabbit artifact.

The historical foundation is intentionally worldwide and layered. The package does not present a single invention narrative: it places ancient Greek, medieval Arabic, classical Chinese, and early-modern European sources beside nineteenth-century psychophysics and modern experimental work. This is a scholarly orientation for separating physical construction, interpretation, and observer evidence; it is not a claim that the current source set exhausts the visual, auditory, or perceptual traditions of any region.

7.2 Visual families

The modern label "visual illusion" sits on a longer experimental history than the package's contemporary review sources alone suggest. Oppel's mid-century catalogue of geometrical-optical illusions is now accessible with translation and commentary [Wade et al., 2017], and the primary nineteenth-century records include Zöllner's account of crossing-line distortions and Müller-Lyer's report of the arrow-wing configuration [Zöllner, 1860, Müller-Lyer, 1889]. These sources are treated as historical anchors, not as evidence that a present-day raster is identical to an original plate or that its observer effect is invariant across conditions.

Gregory's classification is useful as a historical and conceptual orientation, but it is not a universally agreed ontology. DuckRabbit consequently stores visual modality, mechanism, perceptual signature, stimulus requirements, and evidence status as separate facets rather than treating a single label as an explanation [Gregory, 1997]. The visual atlas covers the implemented families currently registered in the package: ambiguous figure, contrast, apparent motion, Müller-Lyer, Poggendorff, Ponzo, Kanizsa-type subjective contour, Ebbinghaus contextual size, and Zöllner/Judd orientation-related geometry. "Coverage" here means one deterministic representative per live entry, not a claim that one raster captures the historical stimulus space.

The duck/rabbit construction is an explicit example of why that distinction matters. Brugger's historical analysis discusses variation among figure variants and observers; DuckRabbit therefore exposes a blend parameter and labels its output as a construction rather than promising a fixed alternation rate or universal bistability [Brugger, 1999]. The Müller-Lyer entry uses typed arrow geometry. Its engineering form is compatible with a family whose image statistics have been analyzed as potentially informative about image-source relationships, but the implementation does not adjudicate that account or infer a perceived-length report [Müller-Lyer, 1889, Howe and Purves, 2005].

The Poggendorff generator makes the occluding geometry and virtual-line orientation explicit. This is appropriate to a literature in which orientation estimation and filtering accounts are theoretically relevant, while avoiding the stronger claim that a particular raster reproduces a published bias without the same observers, viewing conditions, and response task [Morgan, 1999]. The Ponzo construction is similarly bounded: the converging context and test bars are deterministic, whereas the literature contains multiple explanations of Ponzo-like effects rather than one settled mechanism [Fisher, 1967, Yildiz et al., 2022].

Kanizsa-style subjective contours are represented as an illusory-contour construction with explicit inducer geometry. The historical account motivates the family label, while modern reviews place contour completion within a wider literature on grouping, figure-ground organization, attention, and neural mechanisms [Kanizsa, 1976, Wagemans et al., 2012]. Neither source licenses a claim that the generated mask produces a uniform contour percept across observers. The Ebbinghaus entry controls target and surround geometry; classic work demonstrates that judged size depends on context, contour, and comparison conditions, while later work shows that motion and other parameters can alter the effect [Weintraub,

1979, Mruczek et al., 2015]. DuckRabbit's static default is therefore an engineering baseline rather than a task-identical replication. Zöllner/Judd geometry is likewise retained as a source-linked line arrangement. Zöllner's 1860 report supplies the historical primary anchor, while later work supplies a modern analysis of spatial filtering [Zöllner, 1860, Earle and Maskell, 1995]. Its typed line lengths and orientations make the spatial stimulus auditable, while orientation judgments remain outside the package contract.

Finally, apparent motion is included in the visual coverage panel because its defining engineering object is temporal succession, not merely a static pattern. The package verifies frame order, frame rate, timestamps, and frame-to-frame differences. Wertheimer's foundational work and Sekuler's later analysis motivate the family distinction; neither source turns a generated frame strip into a universal report of motion [Wertheimer, 1912, Sekuler, 1996].

7.3 Auditory families

The auditory catalog separates harmonic construction, spectral completion, pitch-class context, dichotic channel assignment, and continuity/masking. A Shepard-like signal is represented through additive partials and explicit envelopes, following a historical literature of tone psychology and later work on assimilation to an internalized musical scale [Stumpf, 1883, Shepard and Jordan, 1984]. The canonical buffer exposes sample rate, amplitude bounds, channel count, partial frequencies, and envelope parameters. It does not determine a listener's perceived pitch height or direction.

The missing-fundamental construction removes a low component while retaining harmonically related partials. This makes the spectral condition reproducible; the associated pitch interpretation remains a listener-level question [Zatorre, 2005]. Tritone and octave entries preserve channel structure, phase, frequency relationships, and timing in typed parameters. Deutsch's foundational accounts and Repp's analysis of spectral-envelope and context effects motivate the evidence records, while also making listener and context dependence central limitations [Deutsch, 1986, Repp, 1997, Deutsch, 1974].

Auditory continuity is represented as an interrupted target with a typed masker and gap. Warren's classic perceptual-restoration result and Riecke and colleagues' separation of sensory and decisional contributions justify the family's inclusion, but a generated masker is not evidence that a listener will report an uninterrupted sound [Warren, 1970, Riecke et al., 2011]. The audio atlas therefore reports RMS, peak, spectral centroid, bandwidth, channel layout, and exact sample-level provenance—not continuity, pitch, or stream judgments.

7.4 Audiovisual and temporal binding families

Audiovisual constructions require a shared clock and an explicit sign convention for offsets. The sound-induced-flash entry creates a typed visual event and one or more audio events; the evidence record links it to the primary demonstration and to a review of the broader literature [Shams et al., 2000, Hirst et al., 2020]. The package can verify event timestamps, sample/frame clocks, and declared offsets. It does not infer a reported flash count, and it does not assume that the same temporal window applies across displays, headphones, latencies, or observers.

The spatial ventriloquist construction treats spatial discrepancy as a parameter and records the channel and display requirements. Reviews describe the ventriloquist illusion as a tool for studying multisensory processing, but the review's scope is not a license to claim spatial capture from a file alone [Bruns, 2019]. More general audiovisual accounts describe integration and segregation as a causal-inference problem shaped by temporal regularities and signal reliability [Noppeney and Lee, 2018]. That framework clarifies why a declared offset is an experimental input, not an observer-level outcome. Temporal ventriloquism receives a separate temporal-binding signature rather than being collapsed into spatial capture. Vroomen and de Gelder manipulated sound–flash timing in a flash-lag task and reported timing-dependent changes under those experimental conditions; Hartcher-O'Brien and Alais studied temporal ventriloquism in a purely temporal context [Vroomen and de Gelder, 2004, Hartcher-O'Brien and Alais, 2011]. DuckRabbit implements the stimulus-side timing contract and leaves the observer-side temporal-recalibration estimate to a future study.

McGurk remains `input_required`. The classic speech study motivates the family, but a lawful and reproducible implementation requires checksummed speech and video fixtures, licensing or consent records, a precise preprocessing contract, and an ethical validation protocol [McGurk and MacDonald, 1976]. Cataloguing the gap is more informative than silently substituting an unrelated synthetic voice or claiming that a generic audiovisual mismatch is a McGurk replication.

7.5 From literature to engineering contract

For each entry, the evidence matrix records a source role, exact source-supported claim, engineering basis, limitation, and audit status. The roles distinguish primary demonstration, review or synthesis, theoretical account, engineering basis, limitation, and input gap. This prevents three common category errors: treating a theory as settled mechanism, treating a review as validation of new code, and treating a generator's deterministic output as a participant result.

The package's literature fidelity is therefore best described as "family-linked, parameterized engineering construction." Some defaults are historically motivated; none should be read as a claim of pixel identity unless that identity is separately established. The checked-in scholarship snapshot provides offline structural validation, including citation-key and DOI/URL format checks. The explicit network audit adds resolver observations and metadata matching; reachability alone is not treated as bibliographic verification. This is a reproducible evidence boundary, not a claim that every source is equally accessible or that theoretical disputes have been resolved.

7.6 Limitations and future observer work

The package does not claim clinical validity, universal effect sizes, cross-cultural invariance, perceptual equivalence across displays or headphones, or observer-level truth from objective media metrics. Playback level, gamma and display calibration, viewing distance, refresh rate, stereo separation, room acoustics, audio transducer response, attention, expectation, language, expertise, and task can all matter. Codec behavior and device latency can introduce additional differences even when decoded media facts match within the declared tolerance.

The synthetic observer diagnostic is intentionally not a substitute for a real vision model or human data. It is a serialized, hand-specified feature function with `training_data=none`, `calibration = analytic`, and `human_data=false`; its metamorphic and sensitivity checks test orchestration, not visual consciousness or human discrimination. A future observer study must pre-register the response scale, reference condition, estimand, exclusions, missing-data rule, randomization seed, display and playback controls, and uncertainty procedure. It must also report the participant and item sampling frame rather than importing a model-output probability as an assumed effect size. The observer harness is consequently study-ready scaffolding, not a result set.

No participant data are bundled with DuckRabbit. The data-availability boundary is intentional: canonical artifacts, encoded fixtures where lawful, source-data sidecars, and deterministic synthetic diagnostics are software outputs; observer outcomes require separately governed data collection. The package is authored by Daniel Ari Friedman and is archived under DOI `10.5281/zenodo.21419693`; release metadata and software-citation guidance are generated from the same identity contract as the manuscript.

8 Publication Atlas, Caption Contract, and Scholarship Audit

The publication layer is generated from the same typed package state as the stimuli. It contains 15 scientific figures, 10 machine-readable tables, and a separately provenance-recorded editorial cover. The cover is an illustration of the project's subject and methods aesthetic; it is not a stimulus, a participant result, or evidence of a perceptual effect.

8.1 Claim-level visualization

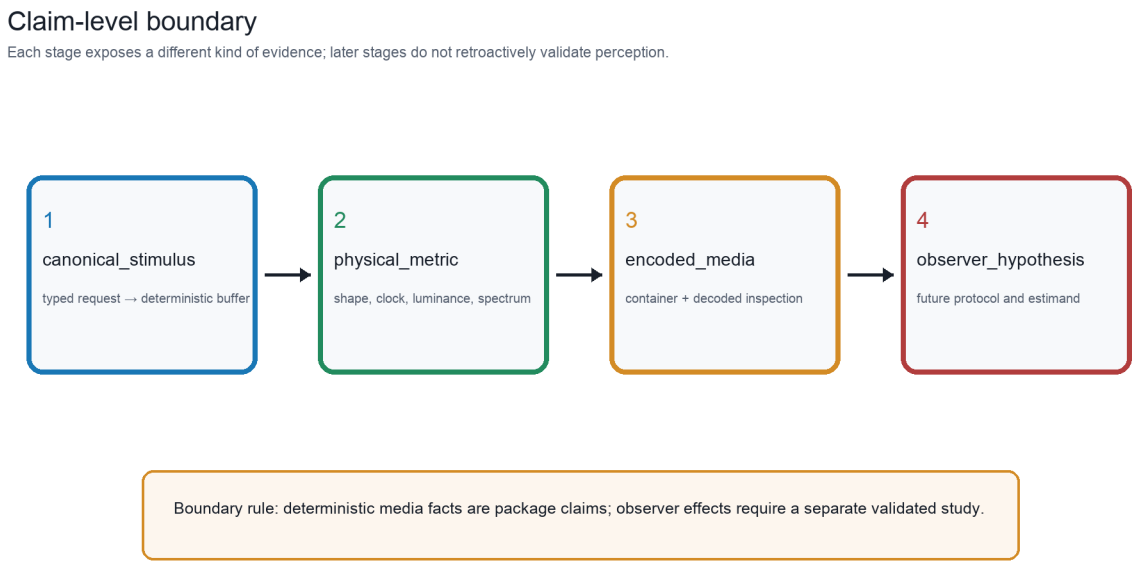


Figure 10: Four numbered boxes separate canonical stimulus, physical metric, encoded media, and observer hypothesis, with a boundary rule stating that observer effects require a separate study.

What this figure shows: Four numbered boxes separate canonical stimulus, physical metric, encoded media, and observer hypothesis, with a boundary rule stating that observer effects require a separate study. The four-stage boundary distinguishes what DuckRabbit can establish directly—canonical stimulus identity, physical/media metrics, and decoded encoded-media facts—from a future observer hypothesis. Source data output/data/claim_boundary.json records the stage definitions and boundary rule; the arrows describe increasing evidential requirements, not an inference that one stage establishes the next. Controls: claim levels in the taxonomy and manifest; seed not applicable to the explanatory diagram. Objective facts: no unit-bearing objective quantity is applicable; stage definitions and evidence boundaries are preserved in the sidecar. Claim level: source_supported. Source data: output/data/claim_boundary.json (SHA-256 digest recorded in the figure registry). Evidence lineage: formalism:eq:observer_estimand, formalism:eq:canonical_digest. Limitations: The diagram is a documentation contract and does not replace a preregistered observer study or empirical data. Boundary: A deterministic artifact can support a future hypothesis but cannot supply the observer data required to test it.

The claim boundary in fig. 10 is a typed epistemic interface. A canonical digest identifies a deterministic buffer; objective metrics identify properties of that buffer; encoded-media verification identifies facts of the delivered file. A future observer hypothesis is a different object with its own protocol, estimand, and uncertainty interval.

8.2 Scholarship map and lineage

What this figure shows: Rows map catalog entries to source tiers, exact supported claims, engineering bases, limitations, and implementation status. The map links every catalog entry to its primary, review, and theory records, then preserves the exact source-supported claim, engineering departure, limitation, and implementation status. Source data output/data/scholarship_map.json are generated from the checked-in evidence matrix; a source record supports only the statement written in that record and does not certify pixel- or task-identical replication. Controls: checked-in evidence matrix; source roles and entry statuses; audit date recorded in data/evidence_matrix.json. Objective facts: one evidence row per catalog entry (18 entries); source-tier counts, exact claim text, engineering basis, limitation, and gap

Source-tiered scholarship map

18 catalog entries; 36 source records; exact claims, engineering bases, and limitations are retained in source data.

catalog entry	primary	review	theory	supported claim	status
visual.duck_rabbit	1	1	0	The entry instantiates a controllable ambiguous-figure stimulus family. limitation recorded: The generator verifies geometry and luminance, not bistable reports.	implemented
visual.simultaneous_contrast	0	1	0	The physical center patches are matched while surround luminance differs. limitation recorded: Display calibration and observer adaptation are not modeled.	implemented
visual.apparent_motion	1	1	0	The sequence contains controlled successive spatial events. limitation recorded: The artifact establishes temporal succession, not perceived motion.	implemented
audio.shepard_tone	1	0	0	The audio contains octave-related partials with controlled sweep parameters. limitation recorded: Playback transducers and listener pitch judgments are external.	implemented
audio.missing_fundamental	0	1	0	The nominal fundamental component is absent from the canonical spectrum. limitation recorded: Pitch completion is an observer-level inference.	implemented
audiovisual.sound_induced_flash	1	1	0	The declared beep/flash event counts and offsets are physically encoded. limitation recorded: The stimulus does not establish a reported flash count.	implemented
audiovisual.ventriloquist	0	2	0	The audio and visual channels carry a declared spatial discrepancy. limitation recorded: Perceived localization depends on room, headphones, display, and observer.	implemented
visual.muller_lyer	0	1	1	The two bar lengths are equal in the canonical raster while wing geometry varies. limitation recorded: The chosen normalized geometry is one engineering variant.	implemented
visual.poggendorff	0	1	1	The occluder and diagonal continuation are generated from explicit geometry. limitation recorded: Alignment judgments and orientation filters are not measured.	implemented
visual.ponzo	1	1	0	Target bars are physically equal and rails converge toward a vanishing region. limitation recorded: Perspective interpretation is observer- and display-dependent.	implemented
visual.kanizsa_triangle	1	1	0	The image contains incomplete inducers with no explicitly drawn triangle edge. limitation recorded: Illusory contour completion is not directly measured.	implemented
visual.ebbinghaus	2	0	0	Central target geometry is held equal while contextual circle geometry differs. limitation recorded: Perceived size depends on viewing scale and context.	implemented
visual.zollner	2	1	0	The entry instantiates a controlled crossing-line orientation stimulus family. limitation recorded: The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction.	implemented
audio.tritone_paradox	2	0	0	The canonical pair has an explicit half-octave frequency relation. limitation recorded: Ascending/descending reports vary across listeners and contexts.	implemented
audio.octave_illusion	1	0	0	The two channels receive alternating octave-related tones. limitation recorded: The auditory percept depends on dichotic presentation and listener.	implemented
audio.auditory_continuity	2	0	0	The canonical audio contains a reproducible interruption interval and masker family. limitation recorded: The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory-versus-decisional effect.	implemented
audiovisual.temporal_ventriloquism	2	2	0	Audio and video event timing and declared offset are deterministic and inspectable. limitation recorded: The generated single-event pair is a simplified engineering stimulus.	implemented
audiovisual.mcgurk	1	0	0	The catalog identifies a speech-dependent audiovisual family without claiming implementation. limitation recorded: No fixture, consent record, or perceptual validation contract is bundled.	input_required

Figure 11: Rows map catalog entries to source tiers, exact supported claims, engineering bases, limitations, and implementation status.

status. Claim level: source_supported. Source data: output/data/scholarship_map.json (SHA-256 digest recorded in the figure registry). Evidence lineage: gregory1997visual, brugger1999duckrabbit, yildiz2022ponzo, hirst2020sound, bruns2019ventriloquist. Limitations: The offline snapshot validates citation and lineage structure; live resolver status is a separate audit, and classifications can remain contested. Boundary: Scholarship coverage bounds the package's evidence record and does not establish that any generated stimulus produces a universal percept.

The source map fig. 11 and tbl. 7 make five distinctions explicit: a primary demonstration, a review, a theoretical account, DuckRabbit's engineering basis, and an evidence limitation. Brugger's duck/rabbit study documents variation across figure variants and observers; this supports a careful ambiguity-family record, not a universal bistability claim [Brugger, 1999]. Yildiz et al. review competing explanations of Ponzo-like illusions, so the catalog retains both a geometric construction and an unresolved theoretical boundary [Yildiz et al., 2022]. Repp's tritone analysis likewise motivates listener- and context-sensitive limitations [Repp, 1997].

The audit also treats software and its metadata as part of the scholarly record. FAIR guidance emphasizes machine-actionable provenance and reuse, and software-citation principles emphasize crediting an identifiable version rather than citing an undifferentiated project name [Wilkinson et al., 2016, Smith et al., 2016]. Accordingly, the release includes `CITATION.cff`, CodeMeta, Zenodo metadata, versioned manifests, source-data sidecars, and resolver-linked bibliography entries. These improve discoverability and attribution; they do not increase the evidential level of any perceptual claim.

The public software identity is explicit: DuckRabbit is published and citable at the DuckRabbit public GitHub repository under the authorship of Daniel Ari Friedman, Active Inference Institute. The development bundle is distinct from the already-published v0.5.0 archive. Current source and manuscript corrections require a new versioned deposit; optional template typesetting and page-level validation are separate gates, not implied by a passing package audit.

8.3 Formal traceability and objective metrics

What this figure shows: Rows connect equation labels to their mathematical definition, implementation paths, tests, figure labels, and claim levels. The traceability registry connects the nine numbered equations to symbols, implementation modules, tests, and registered figures. Source data output/data/formalism_traceability.json is the machine-readable crosswalk used by the manuscript; it demonstrates contract coverage and test linkage without converting formal notation into empirical evidence. Controls: formalism registry version; seed not applicable to the traceability diagram. Objective facts: nine equation records with implementation, test, figure, and claim-level fields; no unit-bearing measurement is plotted. Claim level: physical_metric. Source data: output/data/formalism_traceability.json (SHA-256 digest recorded in the figure registry). Evidence lineage: formalism:eq:canonical_digest, formalism:eq:clock_definition, formalism:eq:objective_statistics. Limitations: Traceability records documentation and verification scope; it does not establish the truth of an observer-level theory. Boundary: A linked equation and test can show an implemented contract, not a validated perceptual law.

The formalism registry fig. 12 links equations eqns. 1, 2, 4, 5, 3, 6, 7, 8, 9 to implementation modules, tests, and figures. The contract is intentionally auditable: equation labels point to code-owned records, while empirical claims remain outside the generator's authority.

What this figure shows: Four artifact cards list image, audio, video, and audiovisual metrics with units, computation provenance, and claim level. The dashboard presents representative measurements from image, audio, video, and audiovisual canonical artifacts. Source data output/data/metrics_dashboard.json retains metric name, value, unit, computation version, tolerance, claim level, and canonical digest; luminance is normalized, amplitude is normalized, spectra are in hertz, frame differences are normalized pixel differences, and synchronization is in milliseconds. Controls: default canonical artifacts; metric computation version and tolerance recorded in the source-data sidecar; seed s=0. Objective facts: finite objective values with units: normalized luminance/amplitude, Hz, normalized pixel difference, ms, counts, and durations. Claim level: physical_metric. Source data: output/data/metrics_dashboard.json (SHA-256 digest recorded in the figure registry). Evidence lineage: formalism:eq:objective_statistics, formalism:eq:temporal_spectral_metrics. Limitations: Metrics are media properties and do not encode pitch, size, motion, localization, binding, or other observer interpretations. Boundary: Metric reproducibility is a package property; interpretation as perception requires a separate observer design and data.

The metrics dashboard fig. 13 reports units, computation version, and canonical digests. Luminance statistics are normalized raster properties; RMS and peak are normalized-amplitude properties; spectral values are in Hz; frame deltas are normalized pixel differences; and synchronization offsets are in milliseconds. None is a psychophysical score.

What this figure shows: Horizontal interval markers show typed domains for dimensions, luminance, levels, frequency, sampling, frame rate, synchronization, and spatial discrepancy. The domain map shows the validated ranges for dimensions, luminance, grayscale and quantization levels, frequency, sampling rate, frame rate, synchronization offset, and spatial discrepancy. Source data output/data/parameter_domains.json records the type name, unit, interval, and engineering

Formalism traceability

Equation labels resolve to implementation modules, tests, figures, and claim levels.

label	equation	implementation	tests	figures
eq:typed_request claim: canonical_stimulus	q = (i, θ, s, e)	registry.py, schema.py	test_generators.py	fig:architecture, fig:formalism_traceability
eq:canonical_generation claim: canonical_stimulus	A = G_i(θ; s)	registry.py, generators.py	test_generators.py, test_v03_contracts.py	fig:architecture, fig:visual_panel
eq:canonical_digest claim: physical_metric	c(A) = Serialize_LE,float32(A, shape, clock, units); h_c(A) = H(c(A))	canonical.py, manifest.py	test_v03_contracts.py	fig:architecture, fig:encoding_verification
eq:encoding_verification claim: encoded_media	F = E_e(A), I = D(F), V(F, M) in {pass, fail}	render.py, inspection.py, manifest.py	test_media_and_render.py, test_v03_contracts.py	fig:encoding_verification
eq:clock_definition claim: physical_metric	N = round(f_s T), t_k = k/f_r, Δ_AV = t_audio − t_video	parameters.py, artifacts.py	test_parameters.py, test_media_and_render.py	fig:temporal_sequence, fig:audiovisual_timeline
eq:objective_statistics claim: physical_metric	μ_Y, σ_Y, RMS_X = M(A; units, tolerance, version)	metrics.py, publication.py	test_v04_scholarly.py	fig:metrics_dashboard, fig:audio_signals
eq:temporal_spectral_metrics claim: physical_metric	D_k = mean |Y_(k+1) − Y_k|; centroid(X) = Σ fM_X(f)/max(ΣM_X(f), 1e-12), M_X = channel-mean |rFFT(X)|	metrics.py, artifacts.py	test_v04_scholarly.py	fig:temporal_sequence, fig:metrics_dashboard
eq:observer_estimand claim: observer_hypothesis	ψ = E[Y conditional on condition and protocol]	observer_analysis.py, observer.py	test_v04_scholarly.py	fig:synthetic_psychophysics, fig:observer_protocol
eq:synthetic_observer claim: synthetic_model_output	p_M(c|A_r,A_c) = σ((w·φ(A_c) − w·φ(A_r))/τ)	synthetic_psychophysics.py, metrics.py	test_synthetic_psychophysics.py	fig:synthetic_psychophysics

Figure 12: Rows connect equation labels to their mathematical definition, implementation paths, tests, figure labels, and claim levels.

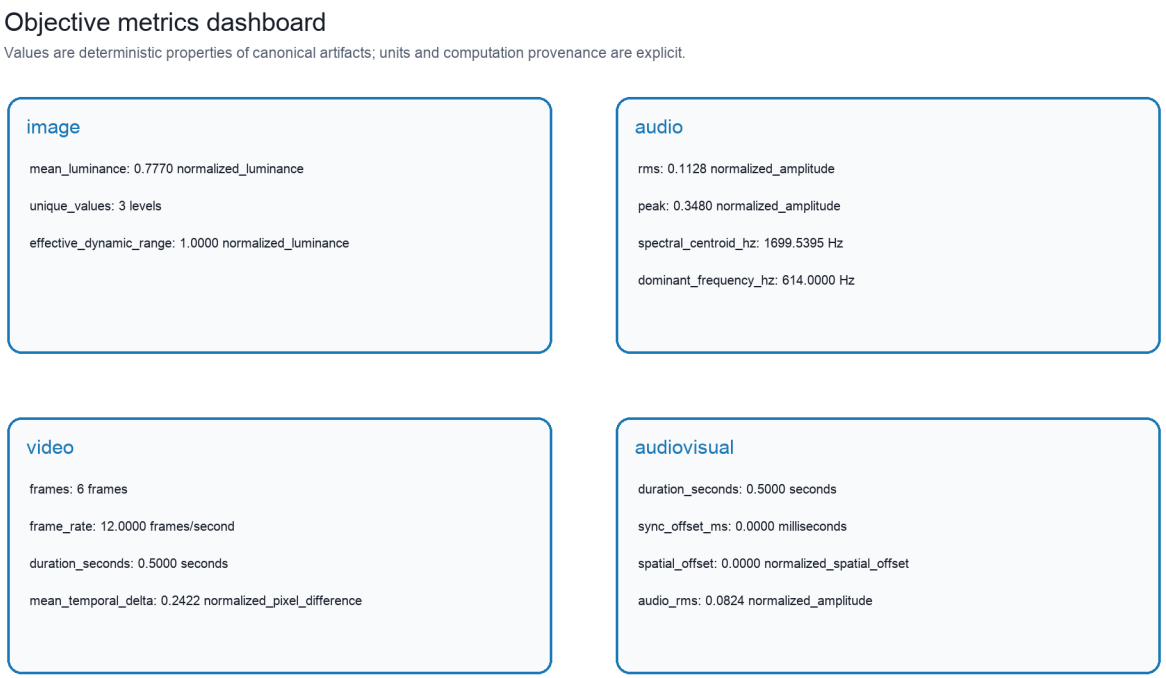


Figure 13: Four artifact cards list image, audio, video, and audiovisual metrics with units, computation provenance, and claim level.



Figure 14: Horizontal interval markers show typed domains for dimensions, luminance, levels, frequency, sampling, frame rate, synchronization, and spatial discrepancy.

role; intervals prevent malformed media and are not proposed as sensitivity thresholds. Controls: parameter schema version and validated scalar domains; seed not applicable to the domain diagram. Objective facts: dimensionless, pixel, level, Hz, samples/s, frames/s, ms, and normalized spatial units as listed per parameter. Claim level: canonical_stimulus. Source data: output/data/parameter_domains.json (SHA-256 digest recorded in the figure registry). Evidence lineage: formalism:eq:typed_request. Limitations: The intervals are software validation bounds and do not encode safe listening levels, display limits, or psychophysical thresholds. Boundary: A valid parameter is a reproducible construction request, not evidence that the requested value is perceptually effective.

The domain map fig. 14 is a validation visualization. Bounds are chosen to prevent malformed media and unsafe encodings, not to predict a viewer, listener, or participant's sensitivity.

8.4 Observer-design boundary

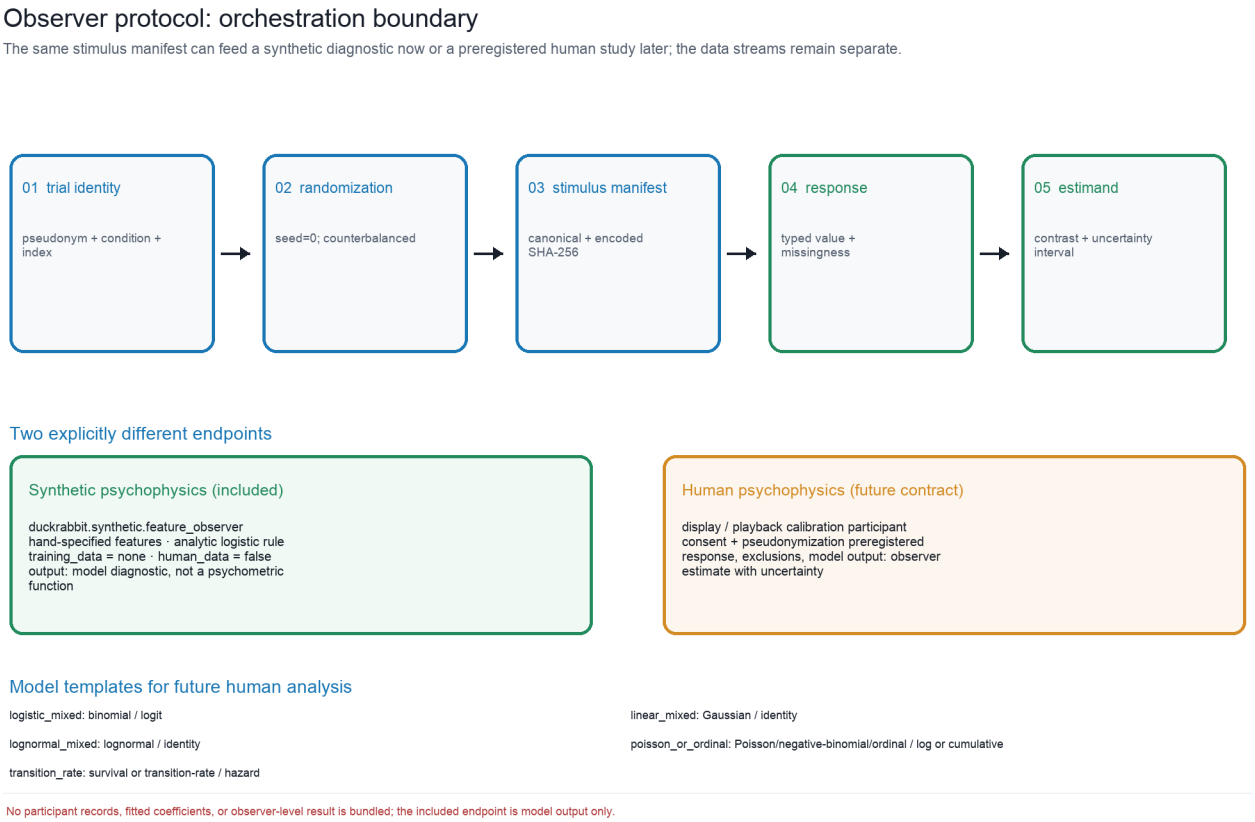


Figure 15: A flow diagram connects trial identity, randomization, stimulus manifest, response, aggregate estimand, and analysis-model templates, with a synthetic-only boundary.

What this figure shows: A flow diagram connects trial identity, randomization, stimulus manifest, response, aggregate estimand, and analysis-model templates, with a synthetic-only boundary. The study-ready scaffold moves from pseudonymous trial identity and deterministic randomization to a stimulus manifest and encoded-file hash, typed response or missingness, and a preregistered estimand. Source data output/data/observer_protocol.json records the study design, randomization seed, model templates, and synthetic-only status; no participant record is included. Controls: study design and deterministic randomization seed recorded in the sidecar; synthetic response generation is separate from human data. Objective facts: trial counts, condition structure, estimand templates, response schemas, and model families; participant outcomes are not applicable. Claim level: observer_hypothesis. Source data: output/data/observer_protocol.json (SHA-256 digest recorded in the figure registry). Evidence lineage: observer:preregistered_estimands. Limitations: The scaffold specifies future data collection and analysis but supplies no participant responses, fitted coefficients, power claim, or validated observer effect. Boundary: The protocol is ready for ethical and preregistered extension, not evidence that the proposed effect exists.

The observer scaffold fig. 15 starts with a pseudonymous trial identity and deterministic randomization, binds the stimulus manifest and encoded hash, records a typed response or missingness state, and ends at a preregistered estimand. The companion diagnostic fig. 9 is a deterministic, hand-specified feature observer with `training_data=none` and `human_data=`

`false`; it is not a pretrained vision model, psychometric function, or participant result. Human psychophysics remains a future evidence layer requiring calibrated playback, consent, preregistration, and observed responses.

8.5 Generated audit tables

Table 6: Code-owned caption, source-data, and accessibility audit.

Figure / claim level	Source and evidence	Controls and objective facts	Limitations, boundary, and accessibility
fig:architecture; canonical stimulus	output/data/architecture.json; formalism:eq:typed request, formalism:eq:canonical generation, formalism:eq:encoding verification	Controls: typed request q=(i,θ,s,e) with seed s=0; default generator and delivery-independent canonicalization Objective: no unit-bearing objective quantity is applicable to this architecture diagram; stage identities and provenance fields are recorded in the sidecar	Limitations: The schematic abstracts implementation boundaries and does not represent an observer study or a causal theory of perception. Boundary: The pipeline verifies reproducible media facts, not what a person experiences. Accessibility: Stage names, mathematical symbols, and arrow direction are printed directly; color is redundant.
fig:catalog matrix; source supported	output/data/catalog matrix.json; gregory1997visual, hirst2020sound, bruns2019ventriloquist	Controls: live taxonomy registry; evidence matrix snapshot; seed not applicable to the catalog diagram Objective: 18 catalog rows; categorical facets and statuses; no observer-level quantity is applicable	Limitations: Taxonomy facets are engineering classifications and remain provisional where the literature supports competing accounts. Boundary: Catalog coverage is bounded to this registered package and does not enumerate all known illusions. Accessibility: Every status and facet is written as text; color only reinforces the printed status.

Figure / claim level	Source and evidence	Controls and objective facts	Limitations, boundary, and accessibility
fig:visual panel; canonical stimulus	output/data/visual panel.json; gregory1997visual, brugger1999duckrabbit, howe2005muller, morgan1999poggendorff, fisher1967ponzo, yildiz2022ponzo, kanizsa1976contours, mruczek2015ebbinghaus, earle1995zollner, wertheimer1912motion, sekuler1996wertheimer	Controls: one default typed parameter object per implemented visual entry; seed s=0; dimensions, luminance bounds, grayscale, and quantization controls remain in each generator schema; temporal entries display their first frame but retain sequence metrics Objective: per tile: mean and standard-deviation relative luminance, unique raster levels, and canonical SHA-256 digest; temporal entries additionally retain frame count, frame rate, and frame deltas; definitions and units are in the source data JSON	Limitations: The gallery is complete for implemented visual entries in this package, not a complete survey of visual illusions; literature families, historical displays, and DuckRabbit rasters are not pixel-identical by default. Viewing scale, display calibration, viewing distance, and observer conditions can change the relevance of a construction; the gallery does not measure an observer's perceptual report. Boundary: The figure establishes coverage of the package's implemented visual constructions and their deterministic raster facts. It does not establish that any viewer will perceive the named signature, nor that the set is exhaustive of the visual-illusion literature. Accessibility: Each tile is labeled by stable illusion ID and accompanied by printed luminance, level-count, and digest fields; color only reinforces grouping and is not required to identify a stimulus.
fig:visual sweep; physical metric	output/data/visual sweep.json; brugger1999duckrabbit, gregory1997visual	Controls: duck weight $\in$ {0,.25,.5,.75,1}; grayscale=quantization $\in$ {2,4,8,16,32}; seed s=0 Objective: 25 canonical rasters; unique levels, normalized mean luminance, and effective dynamic range per cell	Limitations: The grid is a sensitivity of the construction parameters, not a psychophysical sensitivity curve or a validated observer effect. Boundary: The chosen grid is an engineering sampling of the parameter domain and does not imply an optimal or perceptually uniform spacing. Accessibility: Row and column labels state the parameter values; numerical metrics and units are available in the sidecar.

Figure / claim level	Source and evidence	Controls and objective facts	Limitations, boundary, and accessibility
fig:temporal sequence; physical metric	output/data/temporal sequence.json; wertheimer1912motion, sekuler1996wertheimer	Controls: visual.apparent motion default typed parameters; frame rate and frame count from the canonical sequence; seed s=0 Objective: timestamps in seconds, frame rate in frames/s, frame count, rational timebase, and adjacent-frame normalized pixel difference	Limitations: Frame succession and frame deltas are physical file properties; playback timing, display persistence, and motion reports require a controlled observer task. Boundary: The figure documents temporal structure rather than the presence, direction, or strength of perceived motion. Accessibility: Frame index, seconds, frames per second, and normalized-difference labels remain interpretable without color.
fig:audio signals; physical metric	output/data/audio signals.json; shepard1984scale, zatorre2005missing, deutsch1986tritone, repp1997tritone, deutsch1974octave, warren1970continuity, riecke2011continuity	Controls: default typed audio parameters; per-generator channel layout and sample rate; seed s=0; one-sided FFT display with 48 sampled bins Objective: RMS and peak in normalized amplitude; spectral centroid and bandwidth in Hz; sample rate in samples/s; channel layout and canonical SHA-256 digest	Limitations: Playback level, transducer response, dichotic separation, masker design, and listener judgments are external to the canonical buffer. Boundary: The figure compares generated signal properties and literature-linked families; it does not establish pitch, continuity, or stream organization for a listener. Accessibility: Waveform, spectral, amplitude, and frequency labels are printed; bar color is not the sole encoding.
fig:audiovisual timeline; physical metric	output/data/audiovisual timeline.json; shams2000sifi, hirst2020sound, vroomen2004temporal, hartchero-brien2011temporal	Controls: audiovisual.sound induced flash default shared timeline; declared sync offset ΔAV and spatial offset; seed s=0 Objective: video frame-hold luminance and mean-channel absolute audio amplitude on shared time in seconds; ΔAV in ms; spatial discrepancy in normalized units; frame and sample clocks in the sidecar	Limitations: Display refresh, audio hardware, event latency, and observer temporal binding are not measured by this figure. Boundary: A shared-clock construction is not evidence that observers bind the events at the declared or any other perceptual time. Accessibility: The two channels, time axis, signed offset convention, and normalized spatial discrepancy are labeled in text.

Figure / claim level	Source and evidence	Controls and objective facts	Limitations, boundary, and accessibility
fig:encoding verification; encoded media	output/data/encoding verification.json; engineering:typed media adapters	Controls: typed encoding profile selected per artifact; overwrite and backend capability checks; seed inherited from the stimulus request Objective: format availability and verification scope are environment observations; NPZ is exact while other formats are decoded and tolerance-checked	Limitations: Backend availability, codec versions, container metadata, and lossy tolerances are environment-dependent; playback software can expose additional behavior. Boundary: A successful encode or decode does not validate an observer effect or guarantee cross-device perceptual equivalence. Accessibility: Format, backend, preserved fact, and verification columns are textual and remain usable without color.
fig:synthetic psychophysics; synthetic model output	output/data/synthetic psychophysics.json; observer:preregistered estimands	Controls: duck weight comparison against fixed reference; serialized feature weights and temperature; seed s=0 Objective: model probability and score delta on the displayed model-output scale; canonical stimulus digests; human data=false; training data=none	Limitations: The model is hand-specified, has no training data, has not been calibrated against observers, and does not estimate a human threshold or effect size. Its probabilities are model outputs, not participant responses or a psychometric function. Boundary: This diagnostic tests deterministic orchestration and metamorphic input-output behavior only; empirical psychophysics remains future work. Accessibility: Model scale, reference line, model identity, and `human data=false` are printed; color is not needed to interpret the curve.
fig:claim boundary; source supported	output/data/claim boundary.json; formalism:eq:observer estimand, formalism:eq:canonical digest	Controls: claim levels in the taxonomy and manifest; seed not applicable to the explanatory diagram Objective: no unit-bearing objective quantity is applicable; stage definitions and evidence boundaries are preserved in the sidecar	Limitations: The diagram is a documentation contract and does not replace a preregistered observer study or empirical data. Boundary: A deterministic artifact can support a future hypothesis but cannot supply the observer data required to test it. Accessibility: Each stage is named, numbered, and described in text; the boundary rule is readable without color.

Figure / claim level	Source and evidence	Controls and objective facts	Limitations, boundary, and accessibility
fig:scholarship map; source supported	output/data/scholarship map.json; gregory1997visual, brugger1999duckrabbit, yildiz2022ponzo, hirst2020sound, bruns2019ventriloquist	Controls: checked-in evidence matrix; source roles and entry statuses; audit date recorded in data/evidence matrix.json Objective: one evidence row per catalog entry (18 entries); source-tier counts, exact claim text, engineering basis, limitation, and gap status	Limitations: The offline snapshot validates citation and lineage structure; live resolver status is a separate audit, and classifications can remain contested. Boundary: Scholarship coverage bounds the package's evidence record and does not establish that any generated stimulus produces a universal percept. Accessibility: Source roles, claims, limitations, and statuses are printed or available in the sidecar; no category is encoded by color alone.
fig:formalism traceability; physical metric	output/data/formalism traceability.json; formalism:eq:canonical digest, formalism:eq:clock definition, formalism:eq:objective statistics	Controls: formalism registry version; seed not applicable to the traceability diagram Objective: nine equation records with implementation, test, figure, and claim-level fields; no unit-bearing measurement is plotted	Limitations: Traceability records documentation and verification scope; it does not establish the truth of an observer-level theory. Boundary: A linked equation and test can show an implemented contract, not a validated perceptual law. Accessibility: Equation labels, code paths, tests, and figure labels are written as text.
fig:metrics dashboard; physical metric	output/data/metrics dashboard.json; formalism:eq:objective statistics, formalism:eq:temporal spectral metrics	Controls: default canonical artifacts; metric computation version and tolerance recorded in the source-data sidecar; seed s=0 Objective: finite objective values with units: normalized luminance/amplitude, Hz, normalized pixel difference, ms, counts, and durations	Limitations: Metrics are media properties and do not encode pitch, size, motion, localization, binding, or other observer interpretations. Boundary: Metric reproducibility is a package property; interpretation as perception requires a separate observer design and data. Accessibility: Every numerical value is paired with a printed unit or an explicit dimensionless definition.

Figure / claim level	Source and evidence	Controls and objective facts	Limitations, boundary, and accessibility
fig:parameter domains; canonical stimulus	output/data/parameter domains.json; formalism:eq:typed request	Controls: parameter schema version and validated scalar domains; seed not applicable to the domain diagram Objective: dimensionless, pixel, level, Hz, samples/s, frames/s, ms, and normalized spatial units as listed per parameter	Limitations: The intervals are software validation bounds and do not encode safe listening levels, display limits, or psychophysical thresholds. Boundary: A valid parameter is a reproducible construction request, not evidence that the requested value is perceptually effective. Accessibility: Type names, endpoints, units, and roles are printed; interval color is redundant.
fig:observer protocol; observer hypothesis	output/data/observer protocol.json; observer:preregistered estimands	Controls: study design and deterministic randomization seed recorded in the sidecar; synthetic response generation is separate from human data Objective: trial counts, condition structure, estimand templates, response schemas, and model families; participant outcomes are not applicable	Limitations: The scaffold specifies future data collection and analysis but supplies no participant responses, fitted coefficients, power claim, or validated observer effect. Boundary: The protocol is ready for ethical and preregistered extension, not evidence that the proposed effect exists. Accessibility: Every protocol step and the no-participant-data boundary is stated in text; arrows and labels remain legible without color.

Table 7: Source-tiered evidence, DOI/URL verification, exact supported claims, engineering departures, limitations, and explicit planned/input-required gaps.

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
gregory 1997 visual	review; gregory 1997 visual	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	A non-exhaustive classification of visual illusion families.
hirst 2020 sound	review; hirst 2020 sound	DOI: recorded; URL host/path: www.sciencedirect.com	snapshot validated; offline snapshot	A review of sound-induced flash paradigms and multisensory temporal inference.
howe 2005 muller	theory; howe 2005 muller	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	A statistical image-source account of Müller-Lyer geometry.

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
morgan 1999 poggendorff	theory; morgan 1999 poggendorff	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	A mechanistic account of Poggendorff orientation estimation.
fisher 1967 ponzo	primary; fisher 1967 ponzo	DOI: recorded; URL host/path: www.nature.com	snapshot validated; offline snapshot	A primary investigation of identical targets embedded in an angular context.
kanizsa 1976 contours	primary; kanizsa 1976 contours	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Canonical illusory-contour inducer arrangements.
mruczek 2015 ebbinghaus	primary; mruczek 2015 ebbinghaus	DOI: recorded; URL host/path: pmc.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Context-dependent size judgments in Ebbinghaus-family stimuli.
wertheimer 1912 motion	primary; wertheimer 1912 motion	DOI: not recorded; URL host/path: bibbase.org	snapshot validated; offline snapshot	Foundational apparent-motion experiments using successive spatial events.
sekuler 1996 wertheimer	review; sekuler 1996 wertheimer	DOI: recorded; URL host/path: journals.sagepub.com	snapshot validated; offline snapshot	A review connecting Wertheimer's successive-event findings to later apparent-motion research and clarifying their historical scope.
shepard 1984 scale	primary; shepard 1984 scale	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Auditory scale and pitch-class organization in Shepard-like tones.
zatorre 2005 missing	review; zatorre 2005 missing	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	The missing-fundamental problem as a pitch-inference phenomenon.
deutsch 1986 tritone	primary; deutsch 1986 tritone	DOI: recorded; URL host/path: online.ucpress.edu	snapshot validated; offline snapshot	The tritone paradox and dependence on pitch-class context and listener.
deutsch 1974 octave	primary; deutsch 1974 octave	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Dichotic alternating high/low tone construction underlying the octave illusion.
shams 2000 sifi	primary; shams 2000 sifi	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	The one-flash/two-beep sound-induced flash contrast.

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
bruns 2019 ventriloquist	review; bruns 2019 ventriloquist	DOI: recorded; URL host/path: pmc.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Spatial audiovisual capture and multisensory integration constraints.
vroomen 2004 temporal	primary; vroomen 2004 temporal	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Sound timing can alter the apparent temporal structure of a visual event.
mcgurk 1976 speech	primary; mcgurk 1976 speech	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Speech-dependent audiovisual categorical integration, requiring validated fixtures.
warren 1970 continuity	primary; warren 1970 continuity	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Auditory continuity/restoration as a future calibrated masker family.
brugger 1999 duckrabbit	primary; brugger 1999 duckrabbit	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Variation in duck/rabbit ambiguity across figure variants and observers.
yildiz 2022 ponzo	review; yildiz 2022 ponzo	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Competing depth-based and non-depth-based explanations of Ponzo-like illusions.
repp 1997 tritone	primary; repp 1997 tritone	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Listener- and context-dependent effects of spectral envelope and pitch-class structure in tritone judgments.
hartcherobrien 2011 temporal	primary; hartcherobrien 2011 temporal	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Audiovisual timing can shift temporal judgments in a purely temporal context without spatial grounding.
earle 1995 zollner	primary; earle 1995 zollner	DOI: recorded; URL host/path: journals.sagepub.com	snapshot validated; offline snapshot	Crossing oblique and long-line geometry used to measure orientation interactions in the Zöllner family.
zoellner 1860 pseudoscopy	primary; zoellner 1860 pseudoscopy	DOI: recorded; URL host/path: onlinelibrary.wiley.com	snapshot validated; offline snapshot	The nineteenth-century primary description of the crossing-line orientation family later known as the Zöllner illusion.

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
riecke 2011 continuity	primary; riecke 2011 continuity	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Interrupted sounds with masking context and the separation of sensory and decisional contributions.
wagemans 2012 gestalt	review; wagemans 2012 gestalt	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Contemporary perceptual-grouping, contour-completion, and figure-ground accounts relevant to Kanizsa-type constructions.
weintraub 1979 ebbinghaus	primary; weintraub 1979 ebbinghaus	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	A classic contextual-size experiment varying surround geometry and comparison conditions in Ebbinghaus-family displays.
noppeney 2018 causal	review; noppeney 2018 causal	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Causal-inference and temporal-prediction accounts of audiovisual integration and segregation.
alhazen 1989 optics	primary; alhazen 1989 optics	DOI: not recorded; URL host/path: www.cca.qc.ca	snapshot validated; offline snapshot	The English translation and commentary preserves Ibn al-Haytham's eleventh-century Arabic optics as a historical source on direct vision and geometrical image formation.
mozi 2023 canons	review; mozi 2023 canons	DOI: recorded; URL host/path: link.springer.com	snapshot validated; offline snapshot	A modern translation and commentary documents Mohist Canon sections on knowledge, optics, and mechanics, providing a Chinese Warring States-period context for early technical reasoning about vision.

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
ptolemy 1996 optics	primary; ptolemy 1996 optics	DOI: not recorded; URL host/path: sites.dlib.nyu.edu	snapshot validated; offline snapshot	The scholarly translation makes Ptolemy's second-century Optics available as an early work on visual perception and mathematical visual theory.
berkeley 1709 vision	primary; berkeley 1709 vision	DOI: not recorded; URL host/path: www.maths.tcd.ie	snapshot validated; offline snapshot	Berkeley's 1709 essay treats visual distance and space as mediated by learned associations rather than as a simple direct readout of visual geometry.
kircher 1646 light	primary; kircher 1646 light	DOI: not recorded; URL host/path: collections.st-andrews.ac.uk	snapshot validated; offline snapshot	The digitized record anchors Kircher's 1646 Ars magna lucis et umbrae as an early-modern work on light, shadow, and optical display.
molyneux 2020 problem	review; molyneux 2020 problem	DOI: not recorded; URL host/path: plato.stanford.edu	snapshot validated; offline snapshot	The historical account documents Molyneux's 1688 question separating tactile learning from visual recognition and its long cross-sensory afterlife.
cheselden 1728 sight	primary; cheselden 1728 sight	DOI: not recorded; URL host/path: archive.org	snapshot validated; offline snapshot	The digitized Philosophical Transactions record preserves Cheselden's 1728 case report as a historical observation relevant to visual access and the separation of observer evidence from stimulus description.
dai 2015 chinesescience	review; dai 2015 chinesescience	DOI: recorded; URL host/path: link.springer.com	snapshot validated; offline snapshot	A history of Chinese science and technology surveys ancient Chinese physics and records optical phenomena as part of a non-European history of visual science.

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
entry / visual.duck rabbit	engineering / limitation / gap; brugger 1999 duckrabbit; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The entry instantiates a controllable ambiguous-figure stimulus family. Engineering basis: Parameterized ambiguous silhouette; not a pixel-identical reproduction of a historical plate. Limitation: The generator verifies geometry and luminance, not bistable reports. Missing contract: none
entry / visual.simultaneous contrast	engineering / limitation / gap; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The physical center patches are matched while surround luminance differs. Engineering basis: Equal central patches on unequal luminance surrounds. Limitation: Display calibration and observer adaptation are not modeled. Missing contract: none
entry / visual.apparent motion	engineering / limitation / gap; wertheimer 1912 motion; sekuler 1996 wertheimer	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The sequence contains controlled successive spatial events. Engineering basis: Alternating rectangular masks at a fixed rational frame rate. Limitation: The artifact establishes temporal succession, not perceived motion. Missing contract: none

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
entry / audio.shepard tone	engineering / limitation / gap; shepard 1984 scale	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The audio contains octave-related partials with controlled sweep parameters. Engineering basis: Additive octave-spaced partials with a deterministic envelope. Limitation: Playback transducers and listener pitch judgments are external. Missing contract: none
entry / audio.missing fundamental	engineering / limitation / gap; zatorre 2005 missing	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The nominal fundamental component is absent from the canonical spectrum. Engineering basis: Harmonic components are synthesized while the nominal fundamental is omitted. Limitation: Pitch completion is an observer-level inference. Missing contract: none
entry / audiovisual.sound induced flash	engineering / limitation / gap; shams 2000 sifi; hirst 2020 sound	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The declared beep/flash event counts and offsets are physically encoded. Engineering basis: One flash paired with one or two timed beeps on a shared clock. Limitation: The stimulus does not establish a reported flash count. Missing contract: none

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
entry / audiovisual.ventriloquist	engineering / limitation / gap; bruns 2019 ventriloquist; noppeney 2018 causal	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The audio and visual channels carry a declared spatial discrepancy. Engineering basis: Stereo panning and visual displacement encode a spatial discrepancy. Limitation: Perceived localization depends on room, headphones, display, and observer. Missing contract: none
entry / visual.mullerlyer	engineering / limitation / gap; gregory 1997 visual; howe 2005 muller	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The two bar lengths are equal in the canonical raster while wing geometry varies. Engineering basis: Equal bars with controlled arrow-wing geometry. Limitation: The chosen normalized geometry is one engineering variant. Missing contract: none
entry / visual.poggendorff	engineering / limitation / gap; gregory 1997 visual; morgan 1999 poggendorff	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The occluder and diagonal continuation are generated from explicit geometry. Engineering basis: A diagonal continuation is separated by a rectangular occluder. Limitation: Alignment judgments and orientation filters are not measured. Missing contract: none

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
entry / visual.ponzo	engineering / limitation / gap; fisher 1967 ponzo; yildiz 2022 ponzo	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Target bars are physically equal and rails converge toward a vanishing region. Engineering basis: Equal target bars are placed inside converging rails. Limitation: Perspective interpretation is observer- and display-dependent. Missing contract: none
entry / visual.kanizsa triangle	engineering / limitation / gap; kanizsa 1976 contours; wagemans 2012 gestalt	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The image contains incomplete inducers with no explicitly drawn triangle edge. Engineering basis: Three incomplete circular inducers are rasterized around a triangular gap. Limitation: Illusory contour completion is not directly measured. Missing contract: none
entry / visual.ebbinghaus	engineering / limitation / gap; mruczek 2015 ebbinghaus; weintraub 1979 ebbinghaus	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Central target geometry is held equal while contextual circle geometry differs. Engineering basis: Equal central targets are surrounded by different context-circle sizes. Limitation: Perceived size depends on viewing scale and context. Missing contract: none

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
entry / visual.zollner	engineering / limitation / gap; zoellner 1860 pseudoscopy; earle 1995 zollner; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The entry instantiates a controlled crossing-line orientation stimulus family. Engineering basis: Vertical long lines crossed by short oblique inducers with explicit normalized spacing and angle. Limitation: The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction. Missing contract: none
entry / audio.tritone paradox	engineering / limitation / gap; deutsch 1986 tritone; repp 1997 tritone	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The canonical pair has an explicit half-octave frequency relation. Engineering basis: Two octave-complex tones are separated by a half-octave interval. Limitation: Ascending/descending reports vary across listeners and contexts. Missing contract: none
entry / audio.octave illusion	engineering / limitation / gap; deutsch 1974 octave	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The two channels receive alternating octave-related tones. Engineering basis: Alternating high and low tones are assigned to opposite stereo channels. Limitation: The auditory percept depends on dichotic presentation and listener. Missing contract: none

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
entry / audio.auditory continuity	engineering / limitation / gap; warren 1970 continuity; riecke 2011 continuity	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The canonical audio contains a reproducible interruption interval and masker family. Engineering basis: An interrupted sinusoidal carrier is replaced during a declared gap by a deterministic tone or white-noise masker with typed amplitude. Limitation: The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory-versus-decisional effect. Missing contract: none
entry / audiovisual.temporal ventriloquism	engineering / limitation / gap; vroomen 2004 temporal; hartcherobrien 2011 temporal; hirst 2020 sound; noppeney 2018 causal	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Audio and video event timing and declared offset are deterministic and inspectable. Engineering basis: A visual flash and audio click share a clock with explicit offset. Limitation: The generated single-event pair is a simplified engineering stimulus. Missing contract: none

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
entry / audiovisual.mcgurk	engineering / limitation / gap; mcgurk 1976 speech	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The catalog identifies a speech-dependent audiovisual family without claiming implementation. Engineering basis: Requires validated speech audio/video fixtures and licensing records. Limitation: No fixture, consent record, or perceptual validation contract is bundled. Missing contract: A licensed or consented checksummed speech audio/video fixture, synchronization contract, and preregistered perceptual validation protocol are required.

Table 8: Formal equation-to-code traceability.

Label	Definition and symbols	Implementation and tests	Figures and claim level
eq:typed request	q = (i, θ, s, e) Symbols: i, θ, s, e	Code: registry.py, schema.py; Tests: test generators.py	Figures: fig:architecture, fig:formalism traceability; Level: canonical stimulus
eq:canonical generation	A = $G_i(\theta; s)$ Symbols: A, G_i, θ, s	Code: registry.py, generators.py; Tests: test generators.py, test v03 contracts.py	Figures: fig:architecture, fig:visual panel; Level: canonical stimulus
eq:canonical digest	c(A) = Serialize_LE,float32(A, shape, clock, units); h_c(A) = H(c(A)) Symbols: c(A), h_c, H	Code: canonical.py, manifest.py; Tests: test v03 contracts.py	Figures: fig:architecture, fig:encoding verification; Level: physical metric
eq:encoding verification	F = E_e(A), I = D(F), V(F, M) in {pass, fail} Symbols: F, E_e, D, I, V, M	Code: render.py, inspection.py, manifest.py; Tests: test media and render.py, test v03 contracts.py	Figures: fig:encoding verification; Level: encoded media
eq:clock definition	N = round(f_s T), t_k = k/f_r, Δ_AV = t_audio − t_video Symbols: N, f_s, T, t_k, f_r, Δ_AV	Code: parameters.py, artifacts.py; Tests: test parameters.py, test media and render.py	Figures: fig:temporal sequence, fig:audiovisual timeline; Level: physical metric
eq:objective statistics	μ_Y, σ_Y, RMS_X = M(A; units, tolerance, version) Symbols: μ_Y, σ_Y, RMS_X, M	Code: metrics.py, publication.py; Tests: test v04 scholarly.py	Figures: fig:metrics dashboard, fig:audio signals; Level: physical metric

Label	Definition and symbols	Implementation and tests	Figures and claim level
eq:temporal spectral metrics	D_k = mean given Y_(k+1) − Y_k given ; centroid(X) = Σ fM_X(f)/max(ΣM_X(f), 1e-12); M_X = channel-mean given rFFT(X) given Symbols: D_k, M_X, f	Code: metrics.py, artifacts.py; Tests: test v04 scholarly.py	Figures: fig:temporal sequence, fig:metrics dashboard; Level: physical metric
eq:observer estimand	ψ = E[Y conditional on condition and protocol] Symbols: ψ, Y, condition, protocol	Code: observer analysis.py, observer.py; Tests: test v04 scholarly.py	Figures: fig:synthetic psychophysics, fig:observer protocol; Level: observer hypothesis
eq:synthetic observer	p_M(c given A_r,A_c) = σ((w·φ(A_c) − w·φ(A_r))/τ) Symbols: p_M, c, A_r, A_c, w, φ, τ	Code: synthetic psychophysics.py, metrics.py; Tests: test synthetic psychophysics.py	Figures: fig:synthetic psychophysics; Level: synthetic model output

Table 9: Claim ledger separating code-derived facts, scholarship, engineering departures, limitations, scope, and observer hypotheses.

Claim ID	Claim and level	Basis	Lineage and limitation
catalog:count	The catalog contains 18 entries. Level: canonical_stimulus	derived_from_code	Lineage: taxonomy entries() Limitation: Catalog membership is not a completeness claim about all known illusions. Manuscript: manuscript/09 appendix catalog.md
evidence:source count	The checked-in evidence matrix contains 36 source records. Level: source_supported	checked_in_scholarship	Lineage: data/evidence matrix.json::sources Limitation: The offline snapshot does not imply that every URL is currently reachable or that a source supports more than its exact record. Manuscript: manuscript/06 scope and related work.md
scope:no participant data	No participant data are bundled with the package. Level: observer_hypothesis	derived_from_code	Lineage: experiments/observer protocol.md; output/data/synthetic psychophysics.json Limitation: Future observer studies require preregistration, consent, calibrated presentation, and separate data governance. Manuscript: manuscript/07 publication audit.md

Claim ID	Claim and level	Basis	Lineage and limitation
synthetic:diagnostic boundary	The synthetic observer is a deterministic model-output diagnostic, not human psychophysics. Level: synthetic_model_output	synthetic_model_output	Lineage: src/duckrabbit/synthetic psychophysics.py; output/data/synthetic psychophysics.json Limitation: The hand-specified model has no training data and has not been calibrated against observers. Manuscript: manuscript/02 methodology.md
cover:editorial boundary	The cover is a publication illustration, not an experimental stimulus or observer result. Level: publication_illustration	publication_illustration	Lineage: output/reports/cover visualization.json Limitation: The editorial asset is not part of the deterministic scientific stimulus registry. Manuscript: manuscript/07 publication audit.md
evidence:visual.duck rabbit	The entry instantiates a controllable ambiguous-figure stimulus family. Level: source_supported	checked_in_scholarship	Lineage: sources=brugger1999duckrabbit; gregory1997visual; data/evidence matrix.json; engineering=Parameterized ambiguous silhouette; not a pixel-identical reproduction of a historical plate.; limitation=The generator verifies geometry and luminance, not bistable reports.; missing contract=none Limitation: The generator verifies geometry and luminance, not bistable reports. Manuscript: manuscript/06 scope and related work.md
evidence:visual.simultaneous contrast	The physical center patches are matched while surround luminance differs. Level: source_supported	checked_in_scholarship	Lineage: sources=gregory1997visual; data/evidence matrix.json; engineering=Equal central patches on unequal luminance surrounds.; limitation=Display calibration and observer adaptation are not modeled.; missing contract=none Limitation: Display calibration and observer adaptation are not modeled. Manuscript: manuscript/06 scope and related work.md

Claim ID	Claim and level	Basis	Lineage and limitation
evidence:visual.apparent motion	The sequence contains controlled successive spatial events. Level: source_supported	checked_in_scholarship	Lineage: sources=wertheimer1912motion; sekuler1996wertheimer; data/evidence matrix.json; engineering=Alternating rectangular masks at a fixed rational frame rate.; limitation=The artifact establishes temporal succession, not perceived motion.; missing contract=none Limitation: The artifact establishes temporal succession, not perceived motion. Manuscript: manuscript/06 scope and related work.md
evidence:audio.shepard tone	The audio contains octave-related partials with controlled sweep parameters. Level: source_supported	checked_in_scholarship	Lineage: sources=shepard1984scale; data/evidence matrix.json; engineering=Additive octave-spaced partials with a deterministic envelope.; limitation=Playback transducers and listener pitch judgments are external.; missing contract=none Limitation: Playback transducers and listener pitch judgments are external. Manuscript: manuscript/06 scope and related work.md
evidence:audio.missing fundamental	The nominal fundamental component is absent from the canonical spectrum. Level: source_supported	checked_in_scholarship	Lineage: sources=zatorre2005missing; data/evidence matrix.json; engineering=Harmonic components are synthesized while the nominal fundamental is omitted.; limitation=Pitch completion is an observer-level inference.; missing contract=none Limitation: Pitch completion is an observer-level inference. Manuscript: manuscript/06 scope and related work.md

Claim ID	Claim and level	Basis	Lineage and limitation
evidence:audiovisual.sound induced flash	The declared beep/flash event counts and offsets are physically encoded. Level: source_supported	checked_in_scholarship	Lineage: sources=shams2000sifi; hirst2020sound; data/evidence matrix.json; engineering=One flash paired with one or two timed beeps on a shared clock.; limitation=The stimulus does not establish a reported flash count.; missing contract=none Limitation: The stimulus does not establish a reported flash count. Manuscript: manuscript/06 scope and related work.md
evidence:audiovisual.ventriloquist	The audio and visual channels carry a declared spatial discrepancy. Level: source_supported	checked_in_scholarship	Lineage: sources=bruns2019ventriloquist; noppeney2018causal; data/evidence matrix.json; engineering=Stereo panning and visual displacement encode a spatial discrepancy.; limitation=Perceived localization depends on room, headphones, display, and observer.; missing contract=none Limitation: Perceived localization depends on room, headphones, display, and observer. Manuscript: manuscript/06 scope and related work.md
evidence:visual.muller lyer	The two bar lengths are equal in the canonical raster while wing geometry varies. Level: source_supported	checked_in_scholarship	Lineage: sources=gregory1997visual; howe2005muller; data/evidence matrix.json; engineering=Equal bars with controlled arrow-wing geometry.; limitation=The chosen normalized geometry is one engineering variant.; missing contract=none Limitation: The chosen normalized geometry is one engineering variant. Manuscript: manuscript/06 scope and related work.md

Claim ID	Claim and level	Basis	Lineage and limitation
evidence:visual.poggendorff	The occluder and diagonal continuation are generated from explicit geometry. Level: source_supported	checked_in_scholarship	Lineage: sources=gregory1997visual; morgan1999poggendorff; data/evidence matrix.json; engineering=A diagonal continuation is separated by a rectangular occluder.; limitation=Alignment judgments and orientation filters are not measured.; missing contract=none Limitation: Alignment judgments and orientation filters are not measured. Manuscript: manuscript/06 scope and related work.md
evidence:visual.ponzo	Target bars are physically equal and rails converge toward a vanishing region. Level: source_supported	checked_in_scholarship	Lineage: sources=fisher1967ponzo; yildiz2022ponzo; data/evidence matrix.json; engineering=Equal target bars are placed inside converging rails.; limitation=Perspective interpretation is observer- and display-dependent.; missing contract=none Limitation: Perspective interpretation is observer- and display-dependent. Manuscript: manuscript/06 scope and related work.md
evidence:visual.kanizsa triangle	The image contains incomplete inducers with no explicitly drawn triangle edge. Level: source_supported	checked_in_scholarship	Lineage: sources=kanizsa1976contours; wagemans2012gestalt; data/evidence matrix.json; engineering=Three incomplete circular inducers are rasterized around a triangular gap.; limitation=Illusory contour completion is not directly measured.; missing contract=none Limitation: Illusory contour completion is not directly measured. Manuscript: manuscript/06 scope and related work.md

Claim ID	Claim and level	Basis	Lineage and limitation
evidence:visual.ebbinghaus	Central target geometry is held equal while contextual circle geometry differs. Level: source_supported	checked_in_scholarship	Lineage: sources=mruczek2015ebbinghaus; weintraub1979ebbinghaus; data/evidence matrix.json; engineering=Equal central targets are surrounded by different context-circle sizes.; limitation=Perceived size depends on viewing scale and context.; missing contract=none Limitation: Perceived size depends on viewing scale and context. Manuscript: manuscript/06 scope and related work.md
evidence:visual.zollner	The entry instantiates a controlled crossing-line orientation stimulus family. Level: source_supported	checked_in_scholarship	Lineage: sources=zoellner1860pseudoscopy; earle1995zollner; gregory1997visual; data/evidence matrix.json; engineering=Vertical long lines crossed by short oblique inducers with explicit normalized spacing and angle.; limitation=The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction.; missing contract=none Limitation: The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction. Manuscript: manuscript/06 scope and related work.md
evidence:audio.tritone paradox	The canonical pair has an explicit half-octave frequency relation. Level: source_supported	checked_in_scholarship	Lineage: sources=deutsch1986tritone; repp1997tritone; data/evidence matrix.json; engineering=Two octave-complex tones are separated by a half-octave interval.; limitation=Ascending/descending reports vary across listeners and contexts.; missing contract=none Limitation: Ascending/descending reports vary across listeners and contexts. Manuscript: manuscript/06 scope and related work.md

Claim ID	Claim and level	Basis	Lineage and limitation
evidence:audio.octave illusion	The two channels receive alternating octave-related tones. Level: source_supported	checked_in_scholarship	Lineage: sources=deutsch1974octave; data/evidence matrix.json; engineering=Alternating high and low tones are assigned to opposite stereo channels.; limitation=The auditory percept depends on dichotic presentation and listener.; missing contract=none Limitation: The auditory percept depends on dichotic presentation and listener. Manuscript: manuscript/06 scope and related work.md
evidence:audio.auditory continuity	The canonical audio contains a reproducible interruption interval and masker family. Level: source_supported	checked_in_scholarship	Lineage: sources=warren1970continuity; riecke2011continuity; data/evidence matrix.json; engineering=An interrupted sinusoidal carrier is replaced during a declared gap by a deterministic tone or white-noise masker with typed amplitude.; limitation=The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory-versus-decisional effect.; missing contract=none Limitation: The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory-versus-decisional effect. Manuscript: manuscript/06 scope and related work.md

Claim ID	Claim and level	Basis	Lineage and limitation
evidence:audiovisual.temporal ventriloquism	Audio and video event timing and declared offset are deterministic and inspectable. Level: source_supported	checked_in_scholarship	Lineage: sources=vroomen2004temporal; hartchero-brien2011temporal; hirst2020sound; noppeney2018causal; data/evidence matrix.json; engineering=A visual flash and audio click share a clock with explicit offset.; limitation=The generated single-event pair is a simplified engineering stimulus.; missing contract=none Limitation: The generated single-event pair is a simplified engineering stimulus. Manuscript: manuscript/06 scope and related work.md
evidence:audiovisual.mcgurk	The catalog identifies a speech-dependent audiovisual family without claiming implementation. Level: source_supported	checked_in_scholarship	Lineage: sources=mcgurk1976speech; data/evidence matrix.json; engineering=Requires validated speech audio/video fixtures and licensing records.; limitation=No fixture, consent record, or perceptual validation contract is bundled.; missing contract=A licensed or consented checksummed speech audio/video fixture, synchronization contract, and preregistered perceptual validation protocol are required. Limitation: No fixture, consent record, or perceptual validation contract is bundled. Manuscript: manuscript/06 scope and related work.md
publication:figure count	The publication atlas contains 15 registered scientific figures. Level: physical_metric	derived_from_code	Lineage: publication caption specs() Limitation: The count is a package-state fact, not evidence of empirical validity. Manuscript: manuscript/07 publication audit.md

Claim ID	Claim and level	Basis	Lineage and limitation
publication:table count	The publication appendix contains the generated table set defined by the publication table payload registry. Level: physical_metric	derived_from_code	Lineage: publication table payloads() Limitation: Generated tables summarize package state and do not replace source or observer evidence. Manuscript: manuscript/07 publication audit.md

The source-tiered evidence table tbl. 7, caption audit tbl. 6, formalism registry tbl. 8, and claim ledger tbl. 9 are generated sidecars. They provide a publication appendix without duplicating mutable counts or silently converting literature statements into package results.

The publication artifacts have two distinct provenance boundaries. Scientific figures are deterministic PNG renders whose source-data JSON and SHA-256 digests are checked against the code-owned registry. The cover is an editorial charcoal illustration: its selected candidate, prompt fingerprint, source asset digest, dimensions, variants, and provenance boundary are recorded in `output/reports/cover_visualization.json`. Neither boundary is an observer result, and neither substitutes for display calibration or behavioral data.

The publication audit additionally resolves every formalism edge to a real package module, test file, manuscript equation label, and registered figure. Figure source-data sidecars use a versioned schema with figure identity, generator, seed, and nested data payload; the registry verifies this identity and hash independently of the renderer. This makes a visually plausible but stale or relabeled figure fail the same provenance gate as a tampered stimulus.

9 Discussion and Conclusion

DuckRabbit's principal result is a reproducibility boundary rather than a new psychophysical finding. The same typed request can be regenerated, hashed, encoded, decoded, and audited, while the evidence graph and observer layer remain explicit about what the package cannot infer. This makes the atlas useful for stimulus construction, source comparison, and preregistration without treating a rendered image or sound as behavioral evidence.

The scholarly contribution is correspondingly modest but operational. Rather than flattening visual, auditory, temporal, and multisensory families into a single "illusion" label, the package keeps mechanism, perceptual signature, requirements, evidence role, engineering basis, and implementation status separate. A primary demonstration, a review, a theoretical account, and a source describing the DuckRabbit implementation answer different questions. The appendix and source-data sidecars make those distinctions inspectable at the same time as the generated media.

The work is best understood as a research-software artifact with a bounded methods contribution. Its novelty claim is not that it discovers a new illusion or resolves a disputed mechanism; it is that a heterogeneous stimulus catalog can be represented as typed, reproducible, source-linked, and release-audited objects. This framing is consistent with software-citation and FAIR guidance, which treats versioned software, metadata, provenance, and reuse conditions as part of the research record [Smith et al., 2016, Wilkinson et al., 2016, Lamprecht et al., 2020].

The release is authored by Daniel Ari Friedman of the Active Inference Institute and is published and citable at the DuckRabbit public GitHub repository. The v0.5.0 archive is a published versioned record. Subsequent source and manuscript corrections remain unreleased until a new version is explicitly tagged and deposited. A local package audit does not certify that the archived PDF was rebuilt from the current development manuscript.

9.1 Conclusion

DuckRabbit v0.5.0 turns multimodal illusion generation into an auditable typed pipeline. A request produces a canonical artifact, objective metrics, optional delivery files, decoded inspection, and a versioned evidence-bounded manifest. The catalog now records not only what is implemented, but also what the literature supports and what remains unvalidated.

The main scientific contribution is a boundary: deterministic stimulus facts are reproducible software outputs, while perceptual effects are hypotheses that require observer conditions and data. This boundary permits broad generator coverage without overclaiming. Future additions should contribute a typed parameter contract, a literature record, an engineering-fidelity statement, objective invariants, generated documentation, and tests before entering the implemented registry. Promotion is a release decision about software readiness, not a verdict on whether a perceptual phenomenon is real.

The synthetic-psychophysics layer extends this boundary without crossing it. A transparent feature observer makes model inputs, weights, calibration, and output hashes inspectable, so the full orchestration can be tested today. Its small curve is a diagnostic of that specified model, not evidence that humans share its feature map or response function. Empirical observer work remains a separate, ethically and methodologically governed stage.

10 Appendix A. Complete catalog, source tiers, and evidence and implementation boundaries

This appendix is the authoritative rendered snapshot of DuckRabbit's live catalog. It is generated from the taxonomy registry and evidence matrix during the publication build; the table is not hand-maintained. The snapshot records the package's current scope, not a claim to enumerate every illusion described in the literature.

10.1 How to read the matrix

Each row identifies a registered illusion family and separates five questions that are often collapsed in informal catalogs:

1. What modality, mechanism, and signature does the package assign to the construction?
2. What evidence and implementation status does the checked-in record support?
3. Which sources support the exact statement written in the row?
4. What physical claim is supported by the row?
5. What full source role, resolver, DOI status, and claim level are retained in the machine-readable record?

`implemented` means that DuckRabbit has a deterministic generator and a validated canonical artifact contract. It does not mean that the generated stimulus is a pixel-identical reproduction of a historical experiment or that an observer effect has been re-established. `input_required` means that the family remains catalogued but lacks a lawful, checksummed input fixture and/or the validation contract required for deterministic generation. Evidence status and implementation status are deliberately independent.

The source column is a compact citation-key view. The full source role, resolver or DOI status, exact supported claim, engineering departure, and limitation are retained in the machine-readable evidence matrix and in the generated evidence audit tbl. 7. A source supports only the narrow claim recorded for it; a review or theory record is not evidence that a DuckRabbit rendering produces a universal perceptual effect.

Table 10: DuckRabbit catalog entries, source tiers, and evidence/implementation boundaries.

ID	Construction	Evidence / status	Sources	Supported claim
visual.duck rabbit	visual; ambiguity; bistability	literature backed engineering entry; implemented	brugger 1999 duckrabbit; gregory 1997 visual	The entry instantiates a controllable ambiguous-figure stimulus family.
visual.simultaneous contrast	visual; contrast and context; contrast distortion	literature backed engineering entry; implemented	gregory 1997 visual	The physical center patches are matched while surround luminance differs.
visual.apparent motion	visual; temporal motion; illusory motion	literature backed engineering entry; implemented	wertheimer 1912 motion; sekuler 1996 wertheimer	The sequence contains controlled successive spatial events.
audio.shepard tone	auditory; spectral harmonic; continuity	literature backed engineering entry; implemented	shepard 1984 scale	The audio contains octave-related partials with controlled sweep parameters.
audio.missing fundamental	auditory; spectral harmonic; filling in	literature backed engineering entry; implemented	zatorre 2005 missing	The nominal fundamental component is absent from the canonical spectrum.
audiovisual.sound induced flash	audio_visual; crossmodal temporal; fusion or fission	review backed engineering entry; implemented	shams 2000 sifi; hirst 2020 sound	The declared beep/flash event counts and offsets are physically encoded.

ID	Construction	Evidence / status	Sources	Supported claim
audiovisual.ventriloquist	audio_visual; crossmodal spatial; spatial capture	review backed engineering entry; implemented	bruns 2019 ventriloquist; noppeney 2018 causal	The audio and visual channels carry a declared spatial discrepancy.
visual.muller lyer	visual; geometric alignment; geometric distortion	literature backed engineering entry; implemented	gregory 1997 visual; howe 2005 muller	The two bar lengths are equal in the canonical raster while wing geometry varies.
visual.poggendorff	visual; geometric alignment; geometric distortion	literature backed engineering entry; implemented	gregory 1997 visual; morgan 1999 poggendorff	The occluder and diagonal continuation are generated from explicit geometry.
visual.ponzo	visual; geometric alignment; geometric distortion	literature backed engineering entry; implemented	fisher 1967 ponzo; yildiz 2022 ponzo	Target bars are physically equal and rails converge toward a vanishing region.
visual.kanizsa triangle	visual; contrast and context, geometric alignment; filling in	literature backed engineering entry; implemented	kanizsa 1976 contours; wagemans 2012 gestalt	The image contains incomplete inducers with no explicitly drawn triangle edge.
visual.ebbinghaus	visual; contrast and context, geometric alignment; geometric distortion	literature backed engineering entry; implemented	mruczek 2015 ebbinghaus; weintraub 1979 ebbinghaus	Central target geometry is held equal while contextual circle geometry differs.
visual.zollner	visual; geometric alignment; geometric distortion	literature backed engineering entry; implemented	zoellner 1860 pseudoscopy; earle 1995 zollner; gregory 1997 visual	The entry instantiates a controlled crossing-line orientation stimulus family.
audio.tritone paradox	auditory; spectral harmonic; categorical recoding	literature backed engineering entry; implemented	deutsch 1986 tritone; repp 1997 tritone	The canonical pair has an explicit half-octave frequency relation.
audio.octave illusion	auditory; stream segregation; fusion or fission	literature backed engineering entry; implemented	deutsch 1974 octave	The two channels receive alternating octave-related tones.
audio.auditory continuity	auditory; stream segregation; continuity	literature backed engineering entry; implemented	warren 1970 continuity; riecke 2011 continuity	The canonical audio contains a reproducible interruption interval and masker family.
audiovisual.temporal ventriloquism	audio_visual; crossmodal temporal; temporal binding or recalibration	review backed engineering entry; implemented	vroomen 2004 temporal; hartcherobrien 2011 temporal; hirst 2020 sound; noppeney 2018 causal	Audio and video event timing and declared offset are deterministic and inspectable.
audiovisual.mcgurk	audio_visual; speech categorization; categorical recoding	literature backed input dependent entry; input required	mcgurk 1976 speech	The catalog identifies a speech-dependent audiovisual family without claiming implementation.

10.2 Evidence boundary and future expansion

The matrix is intentionally a living registry snapshot. New families may be added when the package can state a typed parameter contract, deterministic canonicalization rule, validation invariant, and evidence boundary. Planned or input-dependent families are not silently promoted because a source exists: promotion requires an implementable contract, reproducible fixtures where needed, and an explicit statement of what remains unvalidated. McGurk therefore remains `input_required` pending a consented or licensed, checksummed speech/ video fixture and a study-ready validation protocol. The candidate-future catalog is documented separately from this live matrix so that absence is not misread as a claim that a phenomenon is unknown or unimportant.

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